

Data Analytics Capstone
Municipal Correlates of Licensed 5G NR Infrastructure in Brazil:
Presence, Intensity, and Digital-Infrastructure Profiles
Final Report

Rony Anderson Spada Pedroso

Walsh College

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Dr. Sridhar Srinivas

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Project Repository and Source Access

The project repository is <https://github.com/ronyspada2025/walsh-msc-capstone>.

Numerical results are recorded in `reports/tables/headline_results.json` and are tied to the analysis execution of September 14, 2026. Appendix A identifies this execution, the input checksum, source and output hashes, software environment, and reproduction commands. The corresponding source package is supplied as `walsh_capstone_update.zip` ; reproduction requires the source version identified by those records. Large raw extracts remain outside version control. Appendix B and `docs/data_provenance_manifest_v6.csv` provide official source families, public access paths, reference-period information, and unresolved historical grain. The executable analysis starts from the preserved municipal CSV. Dataset-level source documentation supports assessment of provenance, while reconstruction of historical acquisition remains limited by the availability of original snapshots.

Abstract

Licensed 5G New Radio (NR) infrastructure is distributed unevenly across Brazil, raising questions about the municipal conditions associated with deployment. The purpose of this quantitative, nonexperimental, cross-sectional study was to use merged official municipal data to classify Brazilian municipalities by whether they have high licensed NR station counts; examine socioeconomic and infrastructure correlates of NR stations per 100,000 residents; identify digital-infrastructure profiles and their association with macro-region; and explore correlates of high Serviço Limitado Privado (SLP) station intensity as a supplementary analysis. Data from the National Telecommunications Agency (Anatel) and the Brazilian Institute of Geography and Statistics (IBGE) covered 5,571 municipalities, with 5,564 complete cases in the common modeling frame. The analysis combined random forest classification, a two-part hurdle framework, ridge regression, k-means clustering, regional association testing, and exploratory logistic models. Structural prediction excluded near-target indicators, and evaluation used held-out observations, repeated cross-validation, geographic sensitivity checks, and bootstrap uncertainty estimates. The high-count classifier achieved a held-out ROC-AUC of .927; discrimination declined to approximately .82 when the target was normalized by population. NR presence was moderately predictable, while continuous per-capita intensity prediction across the full municipal modeling frame remained limited. Population density, fiber infrastructure, and GDP per capita showed modest associations with NR intensity. Two stable infrastructure profiles were associated with macro-region, indicating marked geographic differences in municipal conditions. The supplementary SLP analysis linked higher station intensity to wealth and inherited infrastructure, with weaker predictive performance across states. These findings support a reproducible framework for municipal screening, comparison, and further

investigation. They also indicate that public structural data alone provide limited support for forecasting deployment intensity across the national municipal frame. Licensed NR counts measure infrastructure presence rather than verified Standalone or cloud-native operation, and SLP remains an exploratory private-network-related proxy. The cross-sectional design, incomplete historical source provenance, and validation limitations constrain interpretation. Engineering and investment decisions therefore require additional evidence on local demand, spectrum, costs, operator strategy, and site feasibility; the study does not establish causal effects or regulatory compliance.

Keywords: 5G New Radio, licensed stations, Brazil, municipal infrastructure, Anatel, IBGE, machine learning, digital divide

Abbreviations

Anatel = Agência Nacional de Telecomunicações (National Telecommunications Agency); ANOVA = analysis of variance; ARI = adjusted Rand index; AUC = area under the curve; CI = confidence interval; CV = cross-validation; EDA = exploratory data analysis; ERB = estação rádio base (radio base station); GDP = gross domestic product; IBGE = Instituto Brasileiro de Geografia e Estatística; LTE = Long-Term Evolution; MAE = mean absolute error; NB = negative binomial; NR = New Radio; NSA = non-standalone; OLS = ordinary least squares; PR-AUC = precision–recall AUC; RAN = radio access network; ROC = receiver operating characteristic; RQ = research question; SA = Standalone; SIDRA = Sistema IBGE de Recuperação Automática; SLP = Serviço Limitado Privado; SMP = Serviço Móvel Pessoal; UF = unidade federativa (federative unit).

Introduction

Background and Context

Fifth-generation mobile systems combine radio infrastructure with increasingly software-defined network functions. In 5G Standalone deployments, cloud-native cores can expose programmable interfaces, support network slicing, and host latency-sensitive services; Open RAN specifications similarly enable disaggregated and virtualized radio functions (International Telecommunication Union [ITU], 2017; O-RAN ALLIANCE, 2026; Polese et al., 2023). Experimental platforms show that end-to-end cloud-native 5G operation is technically feasible outside hyperscale environments (Barrachina-Muñoz et al., 2022). Public municipal records, however, observe licensed NR radio stations rather than core architecture. This study therefore measures the geographic footprint of licensed NR infrastructure and does not treat station counts as proof of Standalone or cloud-native operation.

Brazil's 2021 spectrum auction created municipal coverage obligations and a public monitoring context for national 5G expansion (Anatel, 2021). Deployment occurs across a country with pronounced differences in population density, income, existing fiber and mobile infrastructure, and local permitting conditions (IBGE, 2024a; Lei nº 13.116, 2015; OECD, 2020). These differences matter to Anatel and the Ministry of Communications, to national carriers planning radio investment, and to enterprises considering private connectivity. Serviço Limitado Privado (SLP) records provide a supplementary view of private-network-related activity, but SLP encompasses multiple limited-service uses and is not a direct measure of private 5G.

The study asks a deliberately simple question before applying complex methods: can municipal structure identify where licensed NR deployment appears, and can the same structure explain how intensive deployment becomes once it is present? The central empirical tension is

that deployment presence and high station counts are predictable, while continuous per-capita intensity is much harder to explain. Leakage controls, population-normalized targets, geographic validation, and count models are used to establish that contrast rather than to obscure it.

Problem Statement

Licensed 5G NR infrastructure is distributed unevenly across Brazil, yet prior municipal research has focused mainly on fixed broadband, household access, or selected metropolitan areas rather than the complete 5G-era municipal frame (Gozzi et al., 2024; Macedo & Carvalho, 2013; Nishijima et al., 2017). The objective of this study is to classify and explain Y = licensed 5G NR deployment outcomes — high NR station counts, population-normalized NR station density, digital-infrastructure profile membership, and exploratory high SLP station intensity — for all 5,571 Brazilian municipalities (cross-sectional frame, latest available official vintages) using X = population scale, population density, GDP per capita, fiber intensity, LTE access intensity, and macro-region. Success will be evaluated using held-out and state-grouped predictive performance (ROC-AUC, F1, accuracy against explicit references), inferential coefficients with conventional and state-clustered 95% confidence intervals, cluster validity and stability (silhouette, bootstrap adjusted Rand index), and chi-square/Cramér's V evidence for association between profile membership and region.

Purpose of the Study

The purpose of this quantitative, nonexperimental, cross-sectional study was to use merged Anatel and IBGE municipal data to classify Brazilian municipalities by whether they have high licensed 5G New Radio (NR) station counts; examine socioeconomic and infrastructure correlates of NR stations per 100,000 residents; identify digital-infrastructure

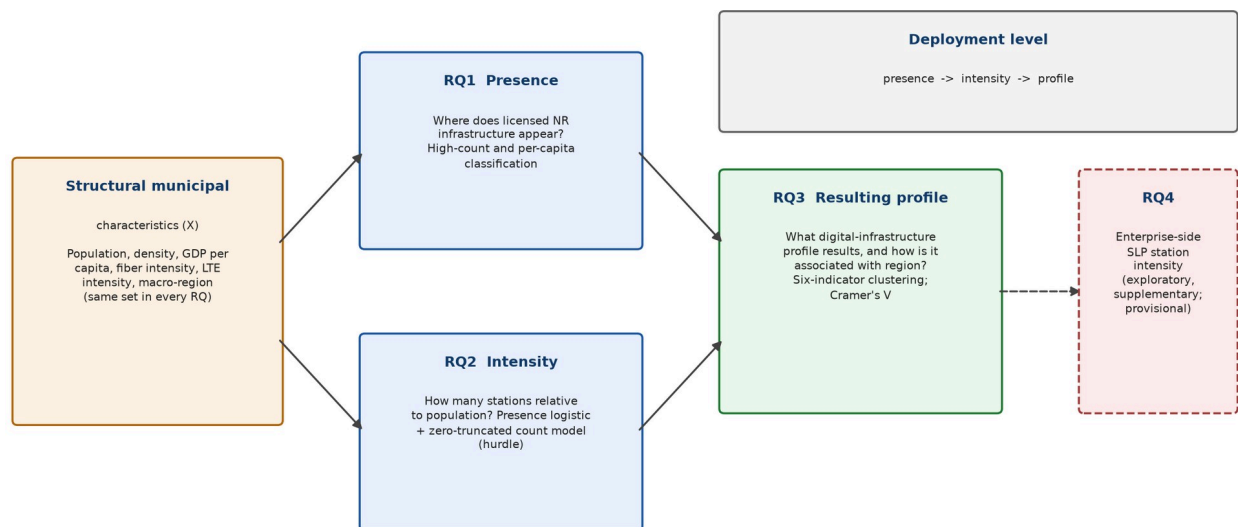
profiles and test their association with macro-region; and explore correlates of high Serviço Limitado Privado (SLP) station intensity as a supplementary analysis.

Research Problems / Research Questions

Figure 1 organizes the study around one core progression: RQ1 asks where substantial NR deployment appears; RQ2 separates whether deployment exists from how intensive it is after arrival; and RQ3 provides supporting geographic context through infrastructure profiles. RQ4 applies the same structural framework to SLP station intensity as an exploratory enterprise-side extension. The confirmatory narrative therefore rests on RQ1–RQ3, with RQ4 reported separately because its historical registry grain remains unverified.

Figure 1

Conceptual Framework: One Research Problem Read at Three Deployment Levels



One research problem: which structural municipal characteristics are associated with the uneven deployment of advanced 5G infrastructure in Brazil, read at three levels (presence, intensity, resulting profile); enterprise private-network demand is an exploratory extension (dashed).

Note. Structural municipal characteristics (orange) are related to licensed NR infrastructure at the presence, intensity, and profile levels (RQ1–RQ3). RQ4 (dashed) is an exploratory extension to enterprise-side SLP station intensity.

Research Question 1 (RQ1 — High-Count NR Rollout Classification): Can structural municipal (socioeconomic and infrastructure) characteristics predict whether a Brazilian

municipality is a high-count licensed 5G NR rollout site, and does the result hold when the target is redefined on a per-capita basis?

Research Question 2 (RQ2 — Structural Correlates of NR Station Intensity): To what extent are structural characteristics associated with the population-normalized intensity of licensed NR station deployment, and which characteristics carry statistically significant associations?

Research Question 3 (RQ3 — Digital-Infrastructure Profile Segmentation): How do municipalities cluster on digital-infrastructure profile indicators, and how strongly is cluster membership associated with geographic region?

Research Question 4 (RQ4 — Correlates of SLP Station Intensity; exploratory, supplementary): Which municipal characteristics are associated with high private-network-related SLP station intensity, and does the association generalize across states?

Three design principles apply across the four questions. First, RQ1 and RQ2 use structural predictors only; NR access and measured download speed are excluded because they are downstream of the deployment process. Second, raw-count results are paired with per-capita outcomes so that municipal scale is not mistaken for infrastructure intensity. Third, predictive performance and coefficient uncertainty are checked for geographic dependence by separating states during validation and clustering inferential standard errors by state. Threshold and SLP aggregation sensitivities are reported as robustness analyses. Table 1 links each question to its variables, method, validation, and output.

Table 1

Research Question–Variable–Method–Output Linkage

RQ	Target and predictors	Method	Validation	Output
RQ1	Target: HIGH_NR_P75 (NR_STATION_CNT > 9); robustness target HIGH_NR_PER_100K_P75 (NR_PER_100K_POP > 46.4). Predictors (structural only): GDP_PER_CAP, POP_2024, POP_DENSITY, FIBER_PER_100, LTE_ACCESS_PER_100, REGION dummies	Random forest; 5-fold CV grid tuning; naive-baseline comparison; P70/P75/P80 sensitivity	Stratified 75/25 split; CV F1; state-grouped 5-fold CV; bootstrap 95% CI for ROC-AUC; per-capita model re-tuned	ROC-AUC with 95% CI, F1 vs. always-positive reference, accuracy vs. majority rate, held-out permutation importance
RQ2	Target: NR_PER_100K_POP (raw, log1p, NR-present conditional) and NR_STATION_CNT with POP_2024 exposure. Same structural predictors	Hurdle model: presence logistic + zero-truncated negative binomial on NR-present municipalities; unconditional NB and RidgeCV as supplementary	75/25 split with bootstrap CI for R ² ; Wald, state-clustered, t(26) and cluster-bootstrap 95% CIs for the count model	Held-out R ² , MAE; presence ROC-AUC with 95% CI; coefficients with conventional and clustered 95% CI
RQ3	Features (log1p, standardized): GDP_PER_CAP, POP_DENSITY, FIBER_PER_100, LTE_ACCESS_PER_100, NR_ACCESS_PER_100, AVG_DL_SPEED	K-means (k = 2–6 by silhouette); chi-square and Cramér's V vs. REGION	Silhouette; 20-resample bootstrap ARI	Infrastructure-profile cluster means, χ^2 , Cramér's V, ARI
RQ4 (exploratory)	Target: HIGH_SLP (SLP_PER_100K_POP > Q75 = 1,618); P70/P80 and aggregation-rule sensitivity; log-intensity OLS. Same structural predictors	Balanced logistic regression; statsmodels LLR test	Stratified 75/25 split; state-grouped 5-fold CV; bootstrap 95% CI for ROC-AUC	ROC-AUC with 95% CI, PR-AUC, precision, sensitivity, specificity, F1, Brier score; odds ratios with conventional and state-clustered 95% CI and p

Note. All variables exist in, or are deterministic transformations of, the supplied merged municipal dataset. REGION is derived from the first digit of the IBGE municipal code.

Contributions and Expected Value

The practical contribution is an all-Brazil municipal screening and benchmarking framework that distinguishes where deployment is likely to appear from how intensive it becomes. The technical contribution is a leakage-audited and population-aware modeling protocol that quantifies two forms of overstatement: near-target predictors raise RQ1 ROC-AUC from .927 to .965, and using raw station counts rather than a per-capita target raises performance from about .82 to about .92. The stakeholder value is bounded and concrete: regulators gain structural context for monitoring, carriers gain a prioritization input, and enterprises gain municipal infrastructure benchmarks. The framework is not a coverage-compliance model and

does not support claims about Standalone cores, virtualized RAN, edge-compute investment, or causal determinants of deployment.

Literature Review

Review Approach and Scope

A structured, targeted review combined peer-reviewed telecommunications and data-science research with official standards, regulatory publications, and Brazilian statistical sources. Searches combined technology terms (5G Standalone, O-RAN, cloud-native 5G, private networks) with economic and geographic terms (deployment cost, investment, population density, digital divide, Brazil, municipality) and methodological terms (hurdle model, negative binomial, spatial cross-validation, leakage, clustering validation). Sources were retained when they informed the study's construct definitions, predictor selection, expected relationships, model choice, or interpretation. Peer-reviewed work was prioritized for empirical and theoretical claims; standards and official publications were used for technical definitions, regulation, and data provenance. Table 2 records the purpose, setting, method, findings, and research-question relevance of each source; the synthesis below develops the theoretical expectations that connect the review to the empirical design.

Table 2

Literature Relevance Matrix

Author (year)	Domain/context	Dataset/setting	Method(s)	Key findings	RQ linkage
Polese et al. (2023)	O-RAN architecture	Systematic survey	Literature review	RAN disaggregation into software-defined, cloud-native functions; NR radios can front non-standalone cores	RQ1–RQ3; footprint caveat
ITU (2017)	IMT-2020 standards	ITU-T Y.3100	Standardization	Terminology and conceptual baseline for 5G networks	RQ1, RQ2
O-RAN ALLIANCE (2026)	Open RAN specs	Specification repository	Standardization	Architecture for virtualized, programmable RAN	RQ1, RQ2

Author (year)	Domain/context	Dataset/setting	Method(s)	Key findings	RQ linkage
Barrachina-Muñoz et al. (2022)	Cloud-native 5G	Experimental platform	Testbed	End-to-end cloud-native 5G with over-the-air transmission is feasible	RQ1, RQ2
Oughton & Frias (2018)	5G deployment economics	Britain, 2020–2030	Techno-economic simulation	Per-capita rollout costs rise steeply in low-density areas; market coverage concentrates where demand concentrates	RQ1–RQ3
Nishijima et al. (2017)	Digital divide, Brazil	National household surveys 2005–2013	Regression, inequality decomposition	Income, education, and region are persistent determinants of digital access	RQ1, RQ3, RQ4
Anatel (2021)	Brazil 5G regulation	2021 spectrum auction	Regulatory documentation	Coverage obligations created a public monitoring trail	RQ1–RQ3
Anatel (2026a, 2026b, 2026c)	Open telecom data	Data panels and open-data files	Administrative records	Official source basis for accesses, stations, SLP, fiber, and speed data	All RQs
IBGE (2024a, 2026a)	Socioeconomic geography	SIDRA municipal aggregates	Official statistics	Population, GDP, area, density, and social-condition indicators	All RQs
Lei nº 13.116 (2015)	Brazilian antenna law	National statute	Legislation	National general rules for licensing, installation, and sharing of telecom infrastructure; municipal implementation remains uneven	RQ1, RQ2; permitting friction
Ericsson (2025)	5G adoption outlook	Global industry data	Industry forecasting	Momentum of 5G SA transition and slow infrastructure ROI	RQ1, RQ2
Nokia (2025)	AI-RAN	Industry partnership	Industry announcement	RAN and AI workloads converging on accelerated cloud infrastructure	RQ1, RQ3
GSMA (2024)	Latin America mobile economy	Regional market data	Industry research	Regional market and enterprise private-network demand context	RQ3, RQ4
Green (1991)	Sample-size guidance	Simulation/rules of thumb	Methodological	$n \geq 50 + 8k$ rule for regression sample sizes	RQ2, RQ4
Cohen (1988)	Statistical power	Benchmark conventions	Methodological	Conventional effect-size benchmarks for power analysis	RQ1, RQ3
Internet Archive (n.d.)	Data acquisition	Wayback Machine	Archival recovery	Recovery of decommissioned data-portal links for reproducibility	Data pipeline
Macedo & Carvalho (2013)	Broadband penetration, Brazil	Brazilian municipalities, 2007 Anatel data	Cross-sectional regression	Penetration concentrated in larger, industrial/service-based municipalities; GDP per capita and population matter	RQ1–RQ3; closest precedent
Abrardi & Cambini (2019)	Broadband investment and adoption	Literature survey	Survey	Investment and adoption respond to demand density, income, and competition	RQ1, RQ2
OECD (2020)	Brazil telecom policy review	Country review	Policy analysis	Municipal permitting and sharing rules are local deployment frictions	RQ1, RQ2
Gozzi et al. (2024)	Digital divide, Brazilian cities	Six metropolises, geolocated speed tests 2017–2023	Spatial statistics	Connectivity improved but its dependence on neighbourhood wealth increased	RQ3; context for RQ1–RQ2

Author (year)	Domain/context	Dataset/setting	Method(s)	Key findings	RQ linkage
OECD (2022, 2025)	Spectrum policy and connectivity divides	OECD countries and Brazil; Latin America	Policy analysis	Coverage obligations attached to about half of 2016–2021 auctions and effective for rural coverage; sharing and neutral-operator models	RQ1, RQ2; policy framing
Bauer & Bohlin (2022)	5G market design and regulation	Conceptual, international	Policy analysis	Coverage obligations substitute for universal-service policy; risk of unreasonable targets under uncertainty	Problem framing; RQ1
Knieps & Bauer (2022)	Economics of 5G local industrial networks	Conceptual	Economic analysis	Local industrial networks as the form of enterprise 5G demand	RQ4 (exploratory)
Breiman (2001)	Random forests	Method paper	Methodological	Ensemble classifier for non-linear, skewed predictors	RQ1
Mullahy (1986); Lambert (1992); Cameron & Trivedi (2013)	Count and hurdle models	Econometric methods	Methodological	Hurdle, zero-inflated, and negative-binomial models for zero-inflated overdispersed counts	RQ2
Anselin (1988)	Spatial econometrics	Methods monograph	Methodological	Formalizes spatial dependence and spatial-lag/error models for geographically structured data	RQ1, RQ2, RQ4; validation design
Roberts et al. (2017)	Spatial cross-validation	Ecological/statistical methods	Methodological	Random splits overstate generalization for spatially structured data; use grouped or blocked CV	RQ1, RQ4
Kaufman et al. (2012)	Data leakage	Data-mining methods	Methodological	Formulation and avoidance of target leakage	RQ1, RQ2
Cramér (1946); Hubert & Arabie (1985); Rousseeuw (1987)	Effect size and cluster validation	Statistical methods	Methodological	Cramér's V, adjusted Rand index, and silhouette as validation measures	RQ3

Note. The matrix connects each source to the four-question design; the thematic synthesis below interprets the sources jointly.

What Previous Research Has Established

The literature supports four connected themes rather than a single deterministic theory of rollout. Deployment economics emphasizes market scale and density; digital-divide research emphasizes income, inherited infrastructure, and regional inequality; standards research defines what NR station records can and cannot measure; and the count-data literature shows why the decision to enter a municipality should be separated from the intensity decision after entry. Together these themes lead directly to the presence-versus-intensity design used in this study.

Deployment Economics and the Presence Margin

Telecommunications investment contains substantial fixed and site-specific costs, so initial rollout tends to favor locations with larger demand pools, greater density, and lower average cost per potential user. Oughton and Frias (2018) model sharply rising per-capita costs in low-density areas, while Abrardi and Cambini (2019) synthesize evidence that broadband investment and adoption respond to demand density, income, and competition. Bauer and Bohlin (2022) and the OECD (2022) show why coverage obligations are used when market incentives alone would leave expensive areas behind. This literature supports the expectation that population scale and density will be strongest at the presence and high-count margin, not necessarily at the conditional intensity margin.

Inherited Infrastructure, Income, and Regional Inequality

Brazilian evidence consistently links connectivity to socioeconomic structure. Macedo and Carvalho (2013) find that municipal fixed-broadband penetration was concentrated in larger, higher-income, industrial and service-oriented municipalities; Nishijima et al. (2017) identify persistent income and regional inequalities in household access; and Gozzi et al. (2024) show that connectivity within six Brazilian cities became more closely associated with neighborhood wealth even as average performance improved. The OECD (2020) further identifies local permitting and infrastructure sharing as geographically variable frictions — frictions that persist even though the Lei Geral das Antenas (Lei nº 13.116, 2015) established national general rules for licensing, installation, and infrastructure sharing, because municipal implementation of the statute remains uneven. These studies support the expectation that income, fiber, legacy LTE access, and region capture accumulated advantages, while also cautioning that these variables are structural correlates rather than exogenous causes.

Why Presence and Intensity Are Different Investment Decisions

The combined literature implies an extensive margin and an intensive margin. A carrier first decides whether a municipality receives any NR footprint; after entry, station density reflects capacity, topology, spectrum, demand, engineering constraints, and local operating choices that are not fully observed in municipal socioeconomic data. This interpretation is an analytical synthesis of the investment and digital-divide literature, and it motivates the hurdle model rather than assuming that one regression should explain both zeros and positive counts. Hurdle and negative-binomial methods are appropriate when outcomes are simultaneously zero-inflated and overdispersed (Cameron & Trivedi, 2013; Mullahy, 1986). The empirical test is therefore not only whether structure predicts deployment, but whether its explanatory power persists after the presence decision has already been made.

Measurement and Policy Tensions

Two tensions define the study's interpretation. First, diffusion accounts anticipate convergence as technology matures, whereas cost-based accounts imply persistent gaps without policy intervention; Brazil's coverage obligations represent an institutional response to that tension (Anatel, 2021; Oughton & Frias, 2018). Second, industry narratives often treat NR expansion as part of cloud-native transformation, while architecture research shows that NR radios can operate with non-standalone cores (Polese et al., 2023). Accordingly, this report treats licensed NR stations as a physical footprint associated with advanced 5G deployment, not as proof of Standalone, cloud-native, or virtualized operation.

Analytical Expectations and Method Selection

Four expectations follow from the synthesis. First, population scale should strongly predict initial deployment and high raw station counts. Second, density, GDP per capita, fiber

intensity, and the legacy LTE base should be positively associated with deployment because they proxy demand and inherited infrastructure. Third, the same variables should explain less of station intensity after NR is already present because that margin depends on unobserved engineering and operator decisions. Fourth, regional inequality should remain visible but should not produce perfectly separated municipal types. Random forests allow nonlinear classification (Breiman, 2001); hurdle and truncated negative-binomial models separate the zero and positive-count processes; state-grouped validation addresses geographic dependence (Roberts et al., 2017), a concern formalized in the spatial-econometrics literature (Anselin, 1988); leakage auditing prevents downstream variables from restating the target (Kaufman et al., 2012); and silhouette, adjusted Rand index, and Cramér's V distinguish cluster stability, separation, and regional association (Cramér, 1946; Hubert & Arabie, 1985; Rousseeuw, 1987).

The Brazilian Municipal Gap and This Study's Contribution

The unresolved empirical gap is a municipal-level, all-of-Brazil account of where licensed NR infrastructure has landed and whether the factors associated with entry also explain intensity. Macedo and Carvalho (2013) provide the closest municipal precedent, but their data predate both 4G and 5G; Nishijima et al. (2017) analyze individuals; Gozzi et al. (2024) analyze selected metropolitan areas; and Oughton and Frias (2018) model another country prospectively. No located study combines licensed NR stations, inherited mobile and fiber infrastructure, and municipal socioeconomics for all 5,571 Brazilian municipalities while auditing leakage and geographic generalization. The contribution is therefore the scale-versus-structure distinction: municipal characteristics predict high station counts strongly, per-capita class moderately, and continuous conditional intensity only weakly.

Materials and Method

Data Sources and Access

All inputs originate from official, open public repositories. IBGE supplies Census 2022 population, territorial area, and density through SIDRA Table 4714 (IBGE, 2022); the 2024 municipal population estimate through Table 6579 (IBGE, 2024b); and municipal GDP through the Produto Interno Bruto dos Municípios series (IBGE, 2025) and SIDRA Table 5938 (IBGE, 2026b). Anatel supplies mobile accesses by technology (Anatel, 2026d), licensed radio stations (Anatel, 2026e), Meu Município fiber/backhaul indicators (Anatel, 2026f), SLP/private-network-related records (Anatel, 2026g), and measured mobile speed (Anatel, 2026h) through its open-data and data-panel infrastructure (Anatel, 2026a, 2026b, 2026c). Table 3 provides a readable source summary; Appendix B provides the complete original public links, current resource names or table identifiers, assurance classes, verification gates, and limitations. The official dataset landing pages and current resources are documented, but the historical raw bytes used to create the frozen merge were not committed, so byte-identical historical reproduction is not claimed.

The analysis frame is the complete official set of 5,571 Brazilian municipalities identified by the seven-digit IBGE code. No municipality is excluded by design, and complete-case modeling is supported by a maximum observed missingness rate of approximately 0.11%. Measured download speed is used only for exploratory analysis and RQ3. Large raw public extracts remain local and git-ignored; the repository publishes source links, provenance metadata, the merged analysis input, and deterministic code rather than duplicating the publishers' full files.

Source Lineage, Assurance, and Merge Design

Provenance has two levels. Computational reproducibility is tied to the preserved municipal input and the recorded software, source, and output hashes in Appendix A. Upstream acquisition is documented at the dataset level: official landing pages and resource or SIDRA identifiers are recorded, with vintage, grain, and unresolved fields stated explicitly. Class A identifies a dataset/table and vintage; Class B identifies the official family and resource but lacks original snapshot bytes or a complete period; Class C retains unresolved historical schema or grain. These classes describe traceability rather than proof of historical extraction. Figure 2 shows the declared source joins, and Table 3 summarizes source-to-variable lineage. The machine-readable manifest is retained at docs/data_provenance_manifest_v6.csv and reproduced in Appendix B.

Table 3

Source-to-Variable Lineage of the Merged Municipal Dataset

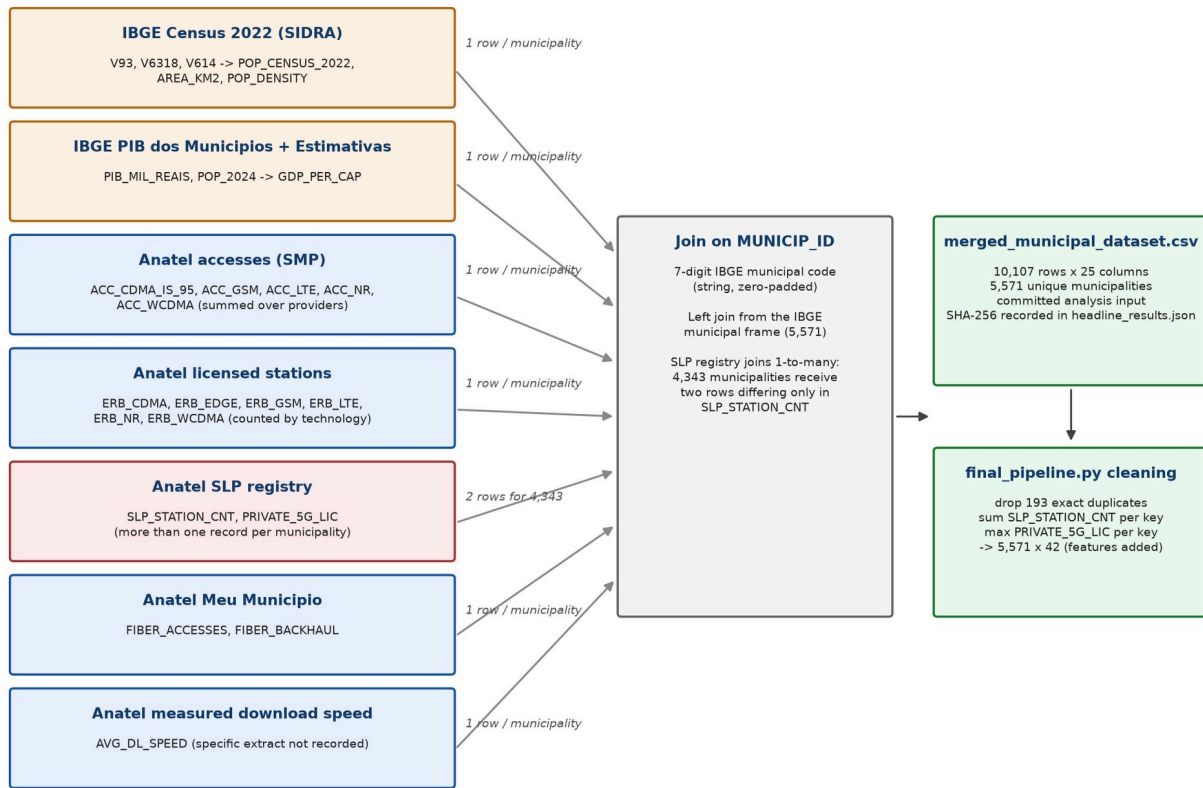
Source family	Official resource	Variables used	Grain after preparation	Vintage / status	Assurance
IBGE Census 2022	SIDRA Table 4714; full URL in Appendix B	POP_CENSUS_2022; AREA_KM2; POP_DENSITY	1 row per municipality	2022 identified	A — exact table/vintage; original export bytes absent
IBGE population estimate	SIDRA Table 6579; full URL in Appendix B	POP_2024	1 row per municipality	2024 identified	A — exact table/vintage; original export bytes absent
IBGE municipal GDP	PIB dos Municípios; SIDRA Table 5938	PIB_MIL_REAIS; GDP_PER_CAP	1 row per municipality	Original extract year unrecorded	B — official series/table; historical vintage incomplete
Anatel mobile accesses (SMP)	Acessos Autorizadas SMP; current ZIP documented	ACC_*; LTE_ACCESS_PER_100; NR_ACCESS_PER_100	Provider-period data aggregated to municipality	Monthly; source month unrecorded	B — official current endpoint; historical ZIP/checksum absent

Source family	Official resource	Variables used	Grain after preparation	Vintage / status	Assurance
Anatel licensed stations	Estações Licenciadas; current ZIP documented	ERB_*; NR_STATION_CNT	Station/technology data aggregated to municipality	Continuous registry; snapshot date unrecorded	B — official current endpoint; historical snapshot absent
Anatel Meu Município	Meu Município panel; current resource name documented	FIBER_ACCESSES; FIBER_BACKHAUL; FIBER_PER_100	Municipality after aggregation	Periodic; source period unrecorded	B — official family/resource; original ZIP/checksum absent
Anatel SLP / Redes Privativas	Official Redes Privativas page; resource candidate documented	SLP_STATION_CNT; PRIVATE_5G_LIC; derived SLP intensity	Two records for 4,343 municipalities in frozen merge	Continuous registry; snapshot date unrecorded	C — source family verified; historical grain unresolved
Anatel measured speed	Official data panels; two candidate resource names	AVG_DL_SPEED	Municipal mean in frozen merge	Periodic; source period unrecorded	C — source family verified; exact historical schema unresolved

Note. All 25 input columns are accounted for; MUNICIPAL_ID is the join key. Table 3 is a concise reader-facing summary. Appendix B gives the complete official URLs, current file names or SIDRA identifiers, verification gates, and limitations. Assurance classes refer to dataset-level identification, not byte-identical historical extraction.

Figure 2

Source-to-Dataset Merge Design



Orange: IBGE sources. Blue: Anatel sources. Red: the one source that joins one-to-many. Grey: join step. Green: committed artifacts. Italic labels give rows per municipality after aggregation.

Note. Orange identifies IBGE sources, blue identifies Anatel sources, red identifies the SLP one-to-many join, grey identifies the seven-digit municipal key, and green identifies committed analysis artifacts. Source assurance and exact public links are reported in Table 3 and Appendix B.

Dataset Structure and Cleaning

The supplied merge contains 10,107 rows, 25 columns, and 5,571 unique municipalities.

Cleaning removes 193 exact duplicates and collapses 4,343 additional municipal-key rows.

Feature engineering produces 42 columns before four binary targets are appended, giving 46 columns afterward. These are not 42 newly engineered predictors. Only SLP_STATION_CNT differs within duplicated pairs. The primary rule sums those counts and takes the maximum SLP presence flag, retaining the supplied information without asserting that the missing historical registry required summation. The original SLP grain is unverified, so RQ4 remains exploratory. Maximum, mean, and

first-record alternatives quantify sensitivity to the rule. Tables 4 and 5 summarize the dataset structure and cleaning decisions.

Table 4

Structural Profile of the Supplied and Cleaned Dataset

Raw rows	Raw columns	Unique municipalities	Duplicate-key rows	Clean rows	Clean columns
10,107	25	5,571	4,343 (after 193 exact duplicates removed)	5,571	42

Note. The 42 columns include original fields and engineered intensities/log1p transforms, before four binary targets (46 total). The classification models use nine structural predictor columns, including four region indicators.

Table 5

Data-Cleaning Log (Reproduced by final_pipeline.py)

Issue	Detection	Treatment	Rationale
Variable-name alignment	Synopsis names absent; equivalent IBGE column names present	Renamed to POP_DENSITY, AREA_KM2, POP_CENSUS_2022, NR_STATION_CNT	Aligns file with the project data dictionary
Exact duplicate rows	193 fully identical rows	Dropped	Duplicates add no information
Duplicate-key merge inflation	9,914 rows vs. 5,571 unique keys (4,343 duplicate rows)	Collapsed per key: first non-null for stable variables; sum for SLP_STATION_CNT; max for PRIVATE_5G_LIC. Alternative SLP rules tested in RQ4.	Preserves both supplied SLP counts without asserting that the unarchived historical registry required summation; RQ4 remains exploratory.
Fiber-flag saturation	100.0% of 5,566 nonmissing FIBER_BACKHAUL values equal 1	Flag retained for traceability; FIBER_PER_100 engineered (median 88.8)	A constant flag cannot discriminate; intensity can
Skewed scales	Heavy right skew and zero inflation (Figures 5 and 6)	log1p transformations and per-capita/per-area intensities engineered	Reduces scale effects; supports transformed visualization of skewed variables

Data Dictionary (Modeled Variables)

Table 6

Data Dictionary for the Modeled Variables

Variable	Definition	Type	Role
MUNICIP_ID	7-digit IBGE municipal code (string)	Key	Join key
POP_2024	Estimated resident population, 2024	Continuous	Predictor (RQ1–RQ4)
POP_DENSITY	Population per km ² , Census 2022	Continuous	Predictor (RQ1–RQ4)

Variable	Definition	Type	Role
GDP_PER_CAP	Municipal GDP per capita (BRL)	Continuous	Predictor (RQ1–RQ4)
FIBER_PER_100	Fiber broadband accesses per 100 inhabitants	Continuous	Predictor (RQ1–RQ4)
LTE_ACCESS_PER_100	LTE accesses per 100 inhabitants	Continuous	Predictor (RQ1–RQ4)
REGION	Macro-region from first digit of MUNICIPAL_ID	Categorical	Predictor dummies; RQ3 association test
NR_STATION_CNT	Licensed NR radio-station count (renamed from ERB_NR); a station-side indirect footprint of advanced 5G deployment that does not establish SA-core or cloud-native operation	Integer	RQ1 target construction; RQ2 count-model outcome
HIGH_NR_P75	1 if NR_STATION_CNT > 9 (75th percentile); a raw-count class that partly encodes municipal scale	Binary	RQ1 primary target (P70/P80 sensitivity)
HIGH_NR_PER_100K_P75	1 if NR_PER_100K_POP > 46.4 (75th percentile)	Binary	RQ1 per-capita robustness target
NR_PER_100K_POP	Licensed NR stations per 100,000 population	Continuous	RQ2 predictive target
NR_ACCESS_PER_100	NR accesses per 100 inhabitants	Continuous	RQ3 feature; excluded from RQ1/RQ2 (leakage)
AVG_DL_SPEED	Mean measured download speed (Mbps)	Continuous	EDA outcome; RQ3 feature; excluded from RQ1/RQ2 (leakage)
SLP_PER_100K_POP	SLP stations per 100,000 population	Continuous	RQ4 target construction
HIGH_SLP	1 if SLP_PER_100K_POP > 1,618 (75th percentile); private-network-related activity, not private 5G specifically	Binary	RQ4 target
PRIVATE_5G_LIC	SLP presence flag (97.8% = 1; saturated); despite its name it records SLP authorization presence, not verified private 5G	Binary	Descriptive only (saturated flag)

Note. Every variable is either a column of the supplied merged dataset or a deterministic transformation of such columns; no external variables were introduced.

Research Questions and Hypotheses

The hypotheses match the statistical procedures applied: predictive claims use held-out metrics and bootstrap intervals, while inferential claims use coefficient intervals or model-level tests. Where a test applies, $\alpha = .05$. State-clustered standard errors account approximately for dependence within 27 federative units; individual coefficient decisions use $t(26)$ -corrected intervals with a state-bootstrap check. Training-only CV, full-frame repeated CV, and held-out estimates are labeled separately. Their comparison describes sensitivity and the observed

train-test gap; it cannot eliminate selection optimism. All findings concern cross-sectional association or prediction, not causation.

RQ1: High-Count NR Rollout Classification

Can structural municipal characteristics classify high-count licensed NR rollout sites, and does discrimination persist for a per-capita target? H0 states that held-out ROC-AUC equals .50; Ha states that it exceeds .50. A 95% percentile interval from 1,000 test-row resamples evaluates discrimination, conditional on the fitted model. The seed-42 development split is supplemented by repeated stratified five-fold CV (five repeats) and a seed-2026 split of the same municipalities. Hyperparameters were selected earlier; these are fixed-configuration sensitivity checks, not nested CV or a test set untouched by model selection. F1 is compared with $2p/(1 + p)$, the always-positive reference (.367 for the count target; .400 for the per-capita target), and accuracy with the majority rate. HIGH_NR_P75 marks NR_STATION_CNT > 9. The per-capita target marks NR_PER_100K_POP above its 75th percentile. It is evaluated both with count-target settings and with a separate training-only grid search, distinguishing a target change from re-tuning. Held-out permutation importance is the mean decrease in ROC-AUC over 20 permutations. It avoids impurity importance's split-selection bias, but correlated features can still affect attribution; neither measure establishes causal importance.

RQ2: Structural Correlates of NR Station Intensity

To what extent are structural characteristics associated with population-normalized NR intensity? A two-part hurdle framework (Cameron & Trivedi, 2013; Mullahy, 1986) separates presence from positive counts. There are 3,377 observed zero NR counts (60.6%) and two unknown counts in the cleaned frame. Part 1 models whether NR is present, using logistic regression. Part 2 fits a zero-truncated negative-binomial model to 2,192 NR-present municipalities. Its log-population offset and centered log-population covariate yield an underlying count-mean parameter proportional to population raised to $1 + \gamma$. The positive-count expectation additionally divides by one minus the underlying zero probability; its elasticity is not simply $1 + \gamma$. Other predictors, including region indicators, are standardized in the estimation frame. An unconditional NB model includes zeros and is supplementary. Predictive H0-P is assessed separately for full-frame raw/log intensity and NR-present-only log intensity, using held-out ridge R^2 and 1,000 test-row bootstrap resamples. A positive interval excluding zero supports predictive value for that specified target and population. These intervals condition on the fitted model and do not quantify geographic transfer or selection uncertainty. Inferential H0-I is tested against an intercept-only truncated model retaining the population offset; individual effects use state-clustered $t(26)$ intervals and a state-bootstrap check. Evaluation uses

held-out $R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$ and $MAE = (1/n) \Sigma|y - \hat{y}|$ for the predictive part, and the truncated-model coefficients, confidence intervals, McFadden pseudo- R^2 , and dispersion parameter for the inferential part.

RQ3: Digital-Infrastructure Profile Segmentation

How do Brazilian municipalities cluster on digital-infrastructure profile indicators, and how strongly is cluster membership associated with geographic region? Null hypothesis (H0): Cluster membership is independent of macro-region. Alternative hypothesis (Ha): Cluster membership and region are associated. The chi-square test of independence, $\chi^2 = \Sigma(O - E)^2 / E$, evaluates the hypothesis; because significance is nearly guaranteed with more than 5,500 observations, the strength of the association is reported with Cramér's $V = \sqrt{(\chi^2 / (n \times \min(r - 1, c - 1)))}$. The silhouette coefficient $s = (b - a) / \max(a, b)$ selects k , and bootstrap adjusted Rand indices establish cluster stability. The six indicators (GDP per capita, density, fiber, LTE, NR access, download speed) describe realized infrastructure and adoption as well as economic capacity, so the clusters are labelled digital-infrastructure profiles rather than “readiness”: a municipality with extensive NR adoption has already experienced part of the phenomenon studied. K-means is used as a transparent, reproducible partitioning of that indicator space; the clusters are descriptive summaries, not a validated latent construct, and they are interpreted only through their indicator profiles.

RQ4: Correlates of SLP Station Intensity (Exploratory, Supplementary)

Which socioeconomic and infrastructure characteristics are associated with the likelihood of high private-network-related SLP station intensity, and does the association generalize across states? RQ4 is designated exploratory and supplementary: its outcome is built by summing two SLP records per municipality under a rule inferred from the join structure, and the raw registry

grain could not be confirmed, so one quarter of the design would otherwise rest on an unverified aggregation. The question is retained because the sensitivity analysis described here shows whether its conclusions depend on that rule, and because the hypothesis can be tested transparently. Null hypothesis (H0): Structural characteristics are jointly unrelated to the likelihood of high SLP intensity (likelihood-ratio test of the logistic model). Alternative hypothesis (Ha): At least one characteristic is significantly associated with that likelihood. The logistic model is reported at the coefficient level — β , standard error, odds ratio per standard deviation, 95% confidence interval, and p — with conventional, state-clustered, t(26)-corrected, and cluster-bootstrap intervals, and a characteristic is described as associated only if the small-cluster-corrected interval excludes an odds ratio of 1. Predictive usefulness is quantified on held-out data by ROC-AUC with a bootstrap 95% confidence interval, precision–recall AUC (reference = prevalence, .25), precision, sensitivity, specificity, F1 (reference = always-positive F1 of .40), accuracy (reference = majority rate of .75), and the Brier score, at the default threshold of .5 on the class-weighted model. Three sensitivity analyses accompany the main estimate: (a) the target is rebuilt with the SLP records aggregated by maximum, mean, and first record instead of sum, and the model re-estimated under each; (b) the top-quartile cut-off is replaced by the 70th and 80th percentiles; and (c) a continuous complement, an OLS model of $\log(1 + \text{SLP stations per } 100,000)$ with state-clustered standard errors, is estimated so that the coefficient-level conclusions do not rest on a dichotomized outcome. The 75th percentile is an analytical choice — it isolates the upper quartile of municipalities as the screening class of interest — not an Anatel or industry threshold, and “high intensity” means above that cut-off as defined in this study. SLP authorizations cover private limited telecommunications services generally; the target measures private-network-related activity rather than private 5G adoption.

Sample-Size Estimation

Because the analysis uses the complete national frame of 5,571 municipalities rather than a sample, nominal statistical power is not the binding constraint of this study. Table 7 records the conventional sample-size checks carried forward from the approved synopsis (Cohen, 1988; Green, 1991) for completeness; every minimum is exceeded by more than an order of magnitude, and the logistic and regression rules of thumb are only loosely applicable to a random forest or a truncated count model. The constraints that matter are practical and structural: geographic dependence among municipalities, the zero inflation and skew of the outcomes, and the indirect-footprint character of the targets, each of which is addressed in the validation protocol and the limitations rather than by sample size.

Table 7

Sample-Size Estimation by Research Question

Research question	Method	Key parameters	Minimum N
RQ1	Power of test (logistic regression)	$\alpha = .05$; power = .80; OR = 1.5 (Cohen, 1988)	N = 317 municipalities
RQ2	Green's rule (k = 9 predictors incl. region dummies)	$n \geq 50 + 8(9)$	N = 122 municipalities
RQ3	Chi-square power test	$\alpha = .05$; power = .80; df = 4; w = 0.30 (Cohen, 1988)	N = 133 municipalities
RQ4	Green's rule (k = 9 predictors incl. region dummies)	$n \geq 50 + 8(9)$	N = 122 municipalities

Note. The analysis frame (N = 5,571; 5,565 complete cases on the structural predictor set, 5,564 on the common modeling frame that also carries the near-target variables used in the leakage demonstration) exceeds each minimum.

Exploratory Data Analysis

Missingness and Descriptive Statistics

Missingness is negligible: Table 8 reports descriptive statistics for the modeled variables and Table 9 lists every variable with observed missingness, profiled in Figure 3. The largest rate is 0.11% (FIBER_PER_100, six municipalities), station counts miss two records, and most

variables miss at most one, supporting complete-case modeling. Distributions are extremely right-skewed and, for NR station counts, zero-inflated: the median municipality has no NR station, while the maximum count is 31,152. In total, 39.3% of municipalities have at least one NR station; the P75 rollout threshold is 9 stations (P70 = 6; P80 = 12), and the mean of the PRIVATE_5G_LIC flag is .978, confirming target saturation.

Table 8

Descriptive Statistics for Modeled Variables (N = 5,571 Municipalities)

Variable	Mean	SD	Median	P75	Max
POP_2024	38,165.8	217,119.5	11,365.5	25,213.8	11,895,578
POP_DENSITY	116.0	595.0	24.3	53.6	13,416.8
GDP_PER_CAP	33,408.0	37,985.3	23,853.8	40,468.7	887,848.8
FIBER_PER_100	98.9	69.0	88.8	147.4	517.7
LTE_ACCESS_PER_100	76.9	35.7	74.5	88.1	1,853.1
NR_ACCESS_PER_100	7.0	13.9	0.0	5.1	129.9
NR_STATION_CNT	44.5	547.8	0.0	9.0	31,152
NR_PER_100K_POP	38.2	105.7	0.0	46.3	3,974.8
AVG_DL_SPEED	43.8	20.5	39.8	52.9	144.2
SLP_PER_100K_POP	4,374.9	24,382.5	577.0	1,618.3	1,028,997.6

Note. Values reproduced by final_pipeline.py from the committed analysis input.

Table 9

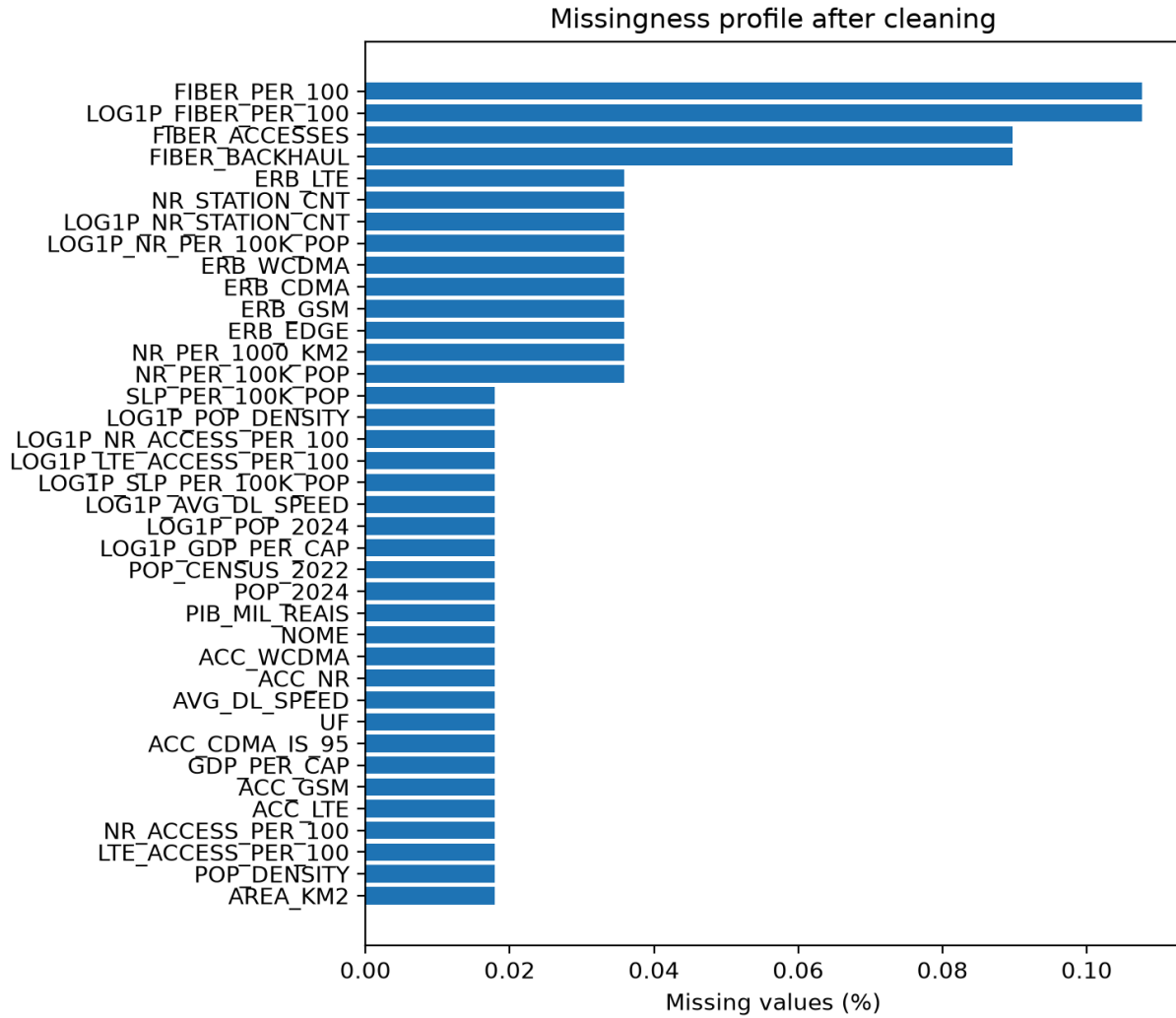
Top Missingness Results After Cleaning

Variable	Missing count	Missing %	Type
FIBER_PER_100 / LOG1P_FIBER_PER_100	6	0.1077	float64
FIBER_ACCESSES / FIBER_BACKHAUL	5	0.0898	float64
ERB_CDMA, ERB_EDGE, ERB_GSM, ERB_LTE, ERB_WCDMA	2	0.0359	float64
NR_STATION_CNT, NR_PER_100K_POP, NR_PER_1000_KM2 (and log1p versions)	2	0.0359	float64

Note. Missingness is very low across available variables; the largest rate is approximately 0.11%, which is not analytically material.

Figure 3

Missingness Profile After Cleaning



Note. Only variables with observed missing values are shown.

Distribution Diagnostics

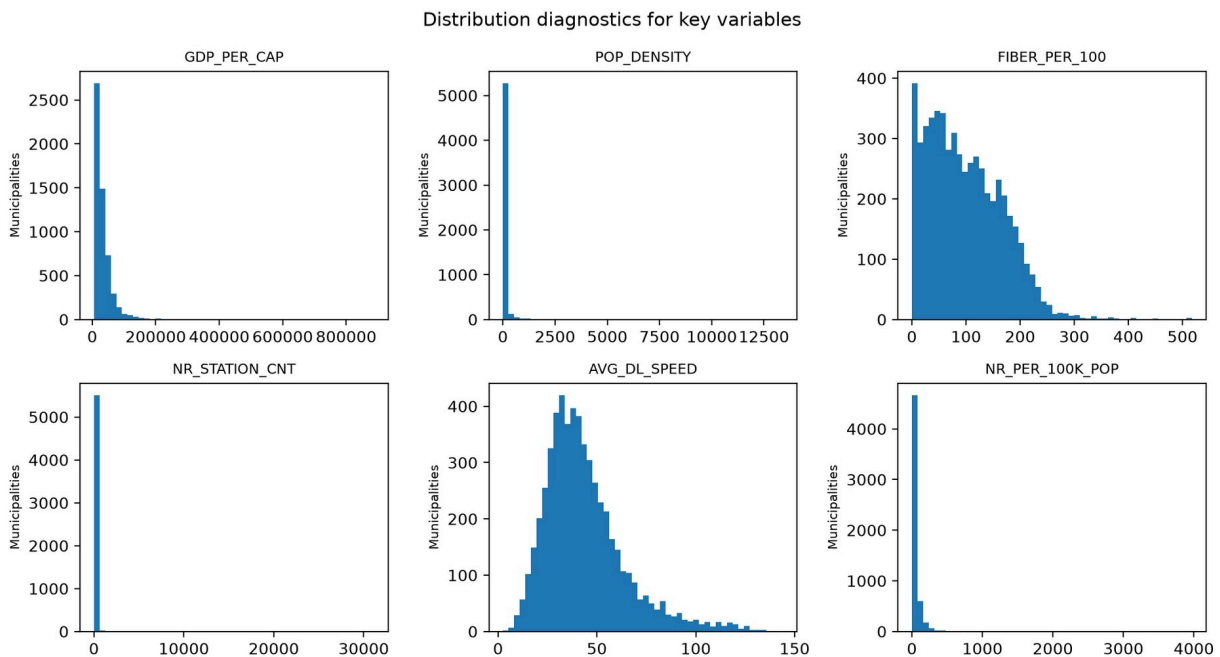
Figure 4 presents raw-scale distribution diagnostics for the key variables. The distributions confirm that Brazilian municipalities differ sharply in demographic scale and infrastructure intensity: median population is approximately 11,366 residents while the maximum exceeds 11.8 million; GDP per capita has a long right tail (median BRL 23,853.78; maximum above BRL 887,000); and 5G NR station counts are especially skewed, which supports percentile-based rollout classes rather than a simple presence/absence definition.

Complementing the raw diagnostics, Figure 5 presents the most skewed variables on raw and

log1p scales side by side: the raw-scale panels are dominated by a single bin, while the log1p panels reveal the underlying structure — approximately log-normal population and GDP per capita, and the pronounced zero spike in NR station counts (60.6% of municipalities have no NR station) that motivates both the percentile-based RQ1 target and the conditional-intensity robustness check in RQ2.

Figure 4

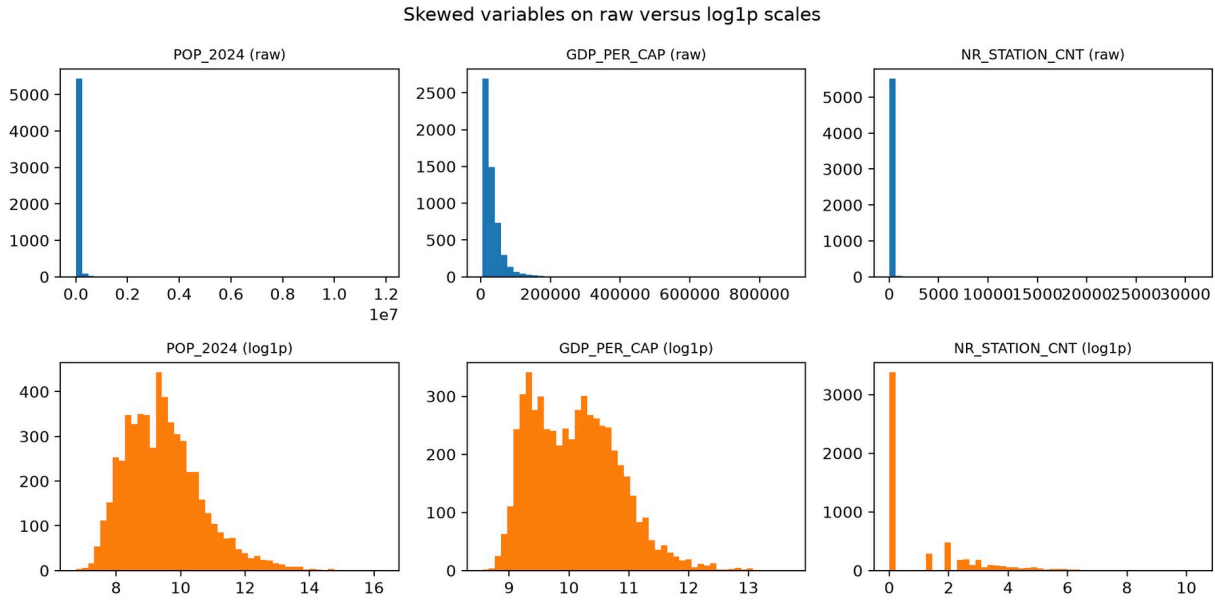
Distribution Diagnostics for Key Variables



Note. The distributions reveal strong right skew in population density, GDP per capita, fiber intensity, NR station counts, and population-normalized station density.

Figure 5

Skewed Variables on Raw Versus log1p Scales



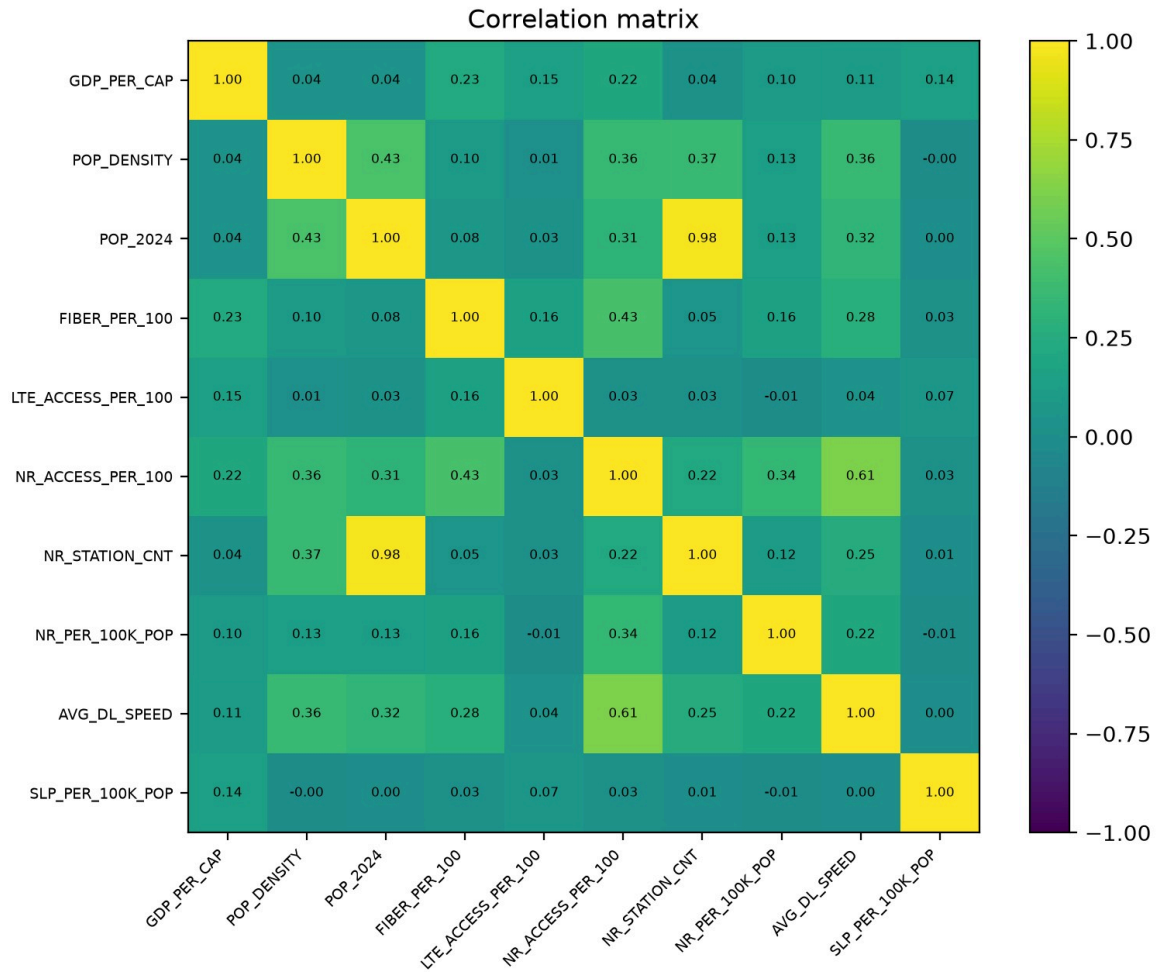
Note. Top row: raw scales. Bottom row: log1p scales. The zero spike in NR station counts reflects the 60.6% of municipalities with no NR station.

Correlation Structure

Figure 6 reports Pearson correlations, interpreted strictly descriptively; the cross-sectional design does not support causal claims. Scale effects dominate the raw counts, which is why modeling uses per-capita intensities. AVG_DL_SPEED correlates most strongly with NR_ACCESS_PER_100 ($r = .61$) and POP_DENSITY ($r = .36$) — an association consistent with, but not evidence of, denser 5G adoption coinciding with better measured performance.

Figure 6

Correlation Matrix for Modeled Variables



Note. Pearson correlations are descriptive and must not be interpreted causally.

Download Speed by Rollout Stage

As supporting evidence, municipalities are grouped into high-count NR ($n = 1,251$; counts > 9), NR present below the P75 threshold ($n = 941$), and a historical no-NR/unknown category ($n = 3,379$). The last group contains 3,377 observed zeros and two missing counts, not 3,379 confirmed absences. Table 10 preserves this declared grouping. Download speeds differ across groups (ANOVA $F = 691.4$, $p < .001$; Figure 7), but the comparison is descriptive and does not identify a causal effect of deployment.

Table 10

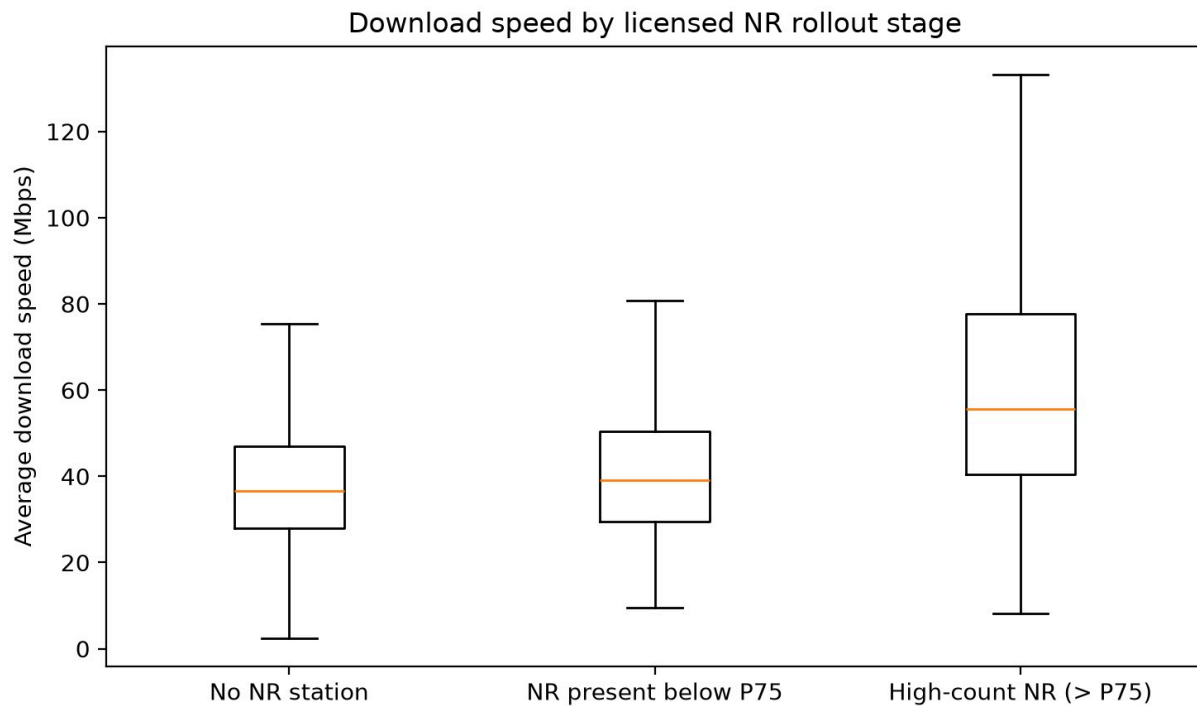
Download Speed by NR Rollout Stage

Rollout stage	n	Mean speed	Median speed	SD	Median fiber	Median NR intensity
High-count NR (> P75)	1,251	60.69	55.63	26.65	140.44	84.62
NR present below P75	941	41.01	39.08	15.58	91.10	34.88
No NR station	3,379	38.35	36.60	14.94	70.15	0.00

Note. Speeds: Mbps; fiber: accesses per 100; NR intensity: stations per 100,000. The No NR station group retains 3,377 recorded zeros plus two unknown counts.

Figure 7

Download Speed by NR Rollout Stage



Note. Boxes show interquartile ranges without outliers.

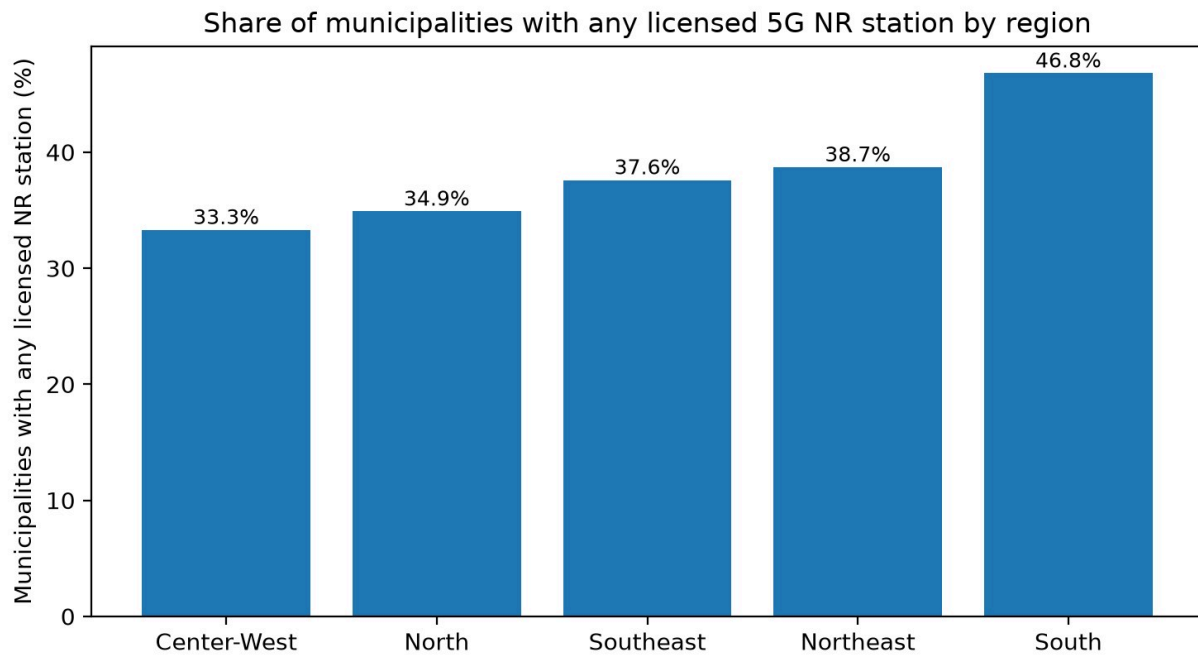
Regional Deployment and Fiber Intensity

Because more than half of municipalities in every region have zero NR footprint, regional medians of station density are uninformative; Figure 8 therefore reports the share of municipalities with any licensed NR deployment: South 46.8%, Northeast 38.7%, Southeast 37.6%, North 34.9%, Center-West 33.3%. Figure 9 plots fiber intensity against station density

with a log-scaled y-axis among NR-present municipalities. Fiber intensity supplies infrastructure context, but this association does not establish that fiber is necessary or sufficient for deployment intensity.

Figure 8

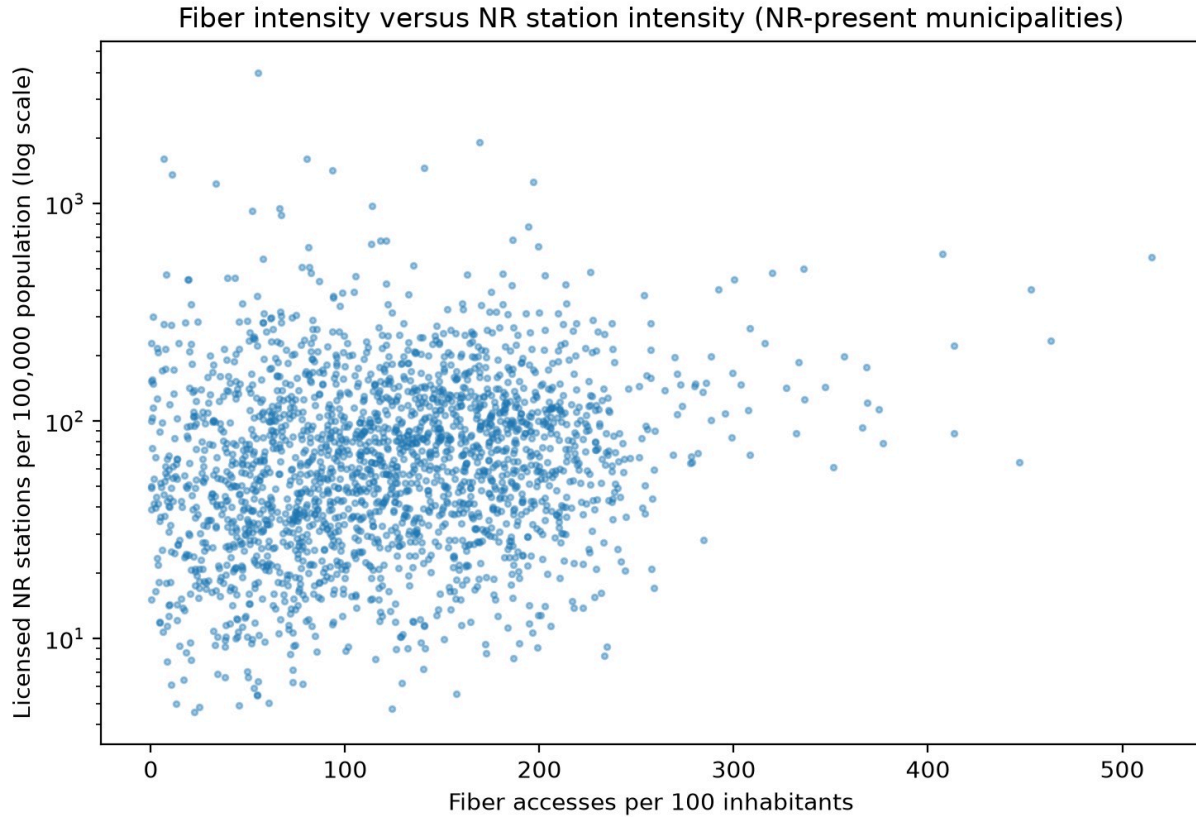
Share of Municipalities With Any 5G NR Deployment by Region



Note. The South has the highest presence share; the Center-West and North the lowest.

Figure 9

Fiber Intensity Versus NR Station Density (Log-Scaled)



Note. Restricted to municipalities with at least one NR station; y-axis log-scaled to expose structure otherwise hidden by skew.

Analytic Approach

Leakage Remediation

A central methodological safeguard of this study concerns predictor–target leakage. Because the high-count NR rollout target is constructed from NR station counts, any predictor that is itself a consequence of NR deployment would inflate apparent performance while explaining nothing. Two available variables have exactly that character: NR_ACCESS_PER_100 is the demand-side trace of the very deployment that defines the target, and AVG_DL_SPEED is downstream of deployed capacity. Both are therefore excluded from the RQ1 and RQ2 predictor sets. The structural predictor set comprises GDP_PER_CAP, POP_2024, POP_DENSITY, FIBER_PER_100, LTE_ACCESS_PER_100, and REGION dummies: socioeconomic scale,

fixed-network intensity, and the legacy mobile base, none of which is mechanically coupled to NR station counts. These are structural municipal characteristics, not exogenous variables in the econometric sense: fiber and LTE intensity are themselves outcomes of earlier telecommunications investment and may share unobserved determinants with NR deployment, which is one reason the results are interpreted as associations rather than causal effects. A naive specification that includes the two near-target variables is estimated once, solely to quantify the optimism leakage produces; it is reported as a methodological demonstration, not as a competing model.

Validation and Inference Protocol

Classifiers use a stratified 75/25 seed-42 development split. RF grid tuning occurs within training data. Repeated stratified 5×5 CV and a seed-2026 split assess a fixed configuration on the same frame; they do not provide nested evaluation or an untouched model-selection holdout. RidgeCV selects the penalty, with scaling fitted before its inner folds. RQ3 uses 20 bootstrap refits. RQ4 separates the balanced predictor from the full-frame inferential logistic model.

ROC-AUC is compared with .50, accuracy with the majority rate, and F1 with the always-positive reference. Training-only CV F1 is $.768 \pm .025$ for the count RF and $.549 \pm .016$ for SLP, compared with held-out F1 .808 and .576. The separate full-frame random-fold SLP value is $.556 \pm .017$ (Table 13). These observed gaps do not rule out selection optimism. State-grouped checks use GroupKFold for count, presence, and SLP models, excluding municipalities from

the same federative unit. Permutation importance is computed on held-out data, and the per-capita RQ1 target is tuned independently as well as evaluated with the count-target settings.

Inferential models report conventional and state-clustered intervals, using $t(26)$ critical values with 27 states. Each state-pairs bootstrap requests 999 draws with seed 42 and fixed transformations. Truncated NB retains 933 draws; SLP retains 898, excluding rank-deficient, nonconverged, or nonfinite fits. Diagnostics record the status of every draw. The analysis was executed on Linux with Python 3.12 and the pinned dependencies listed in Appendix A.

Architecture Diagram/Workflow

System Overview

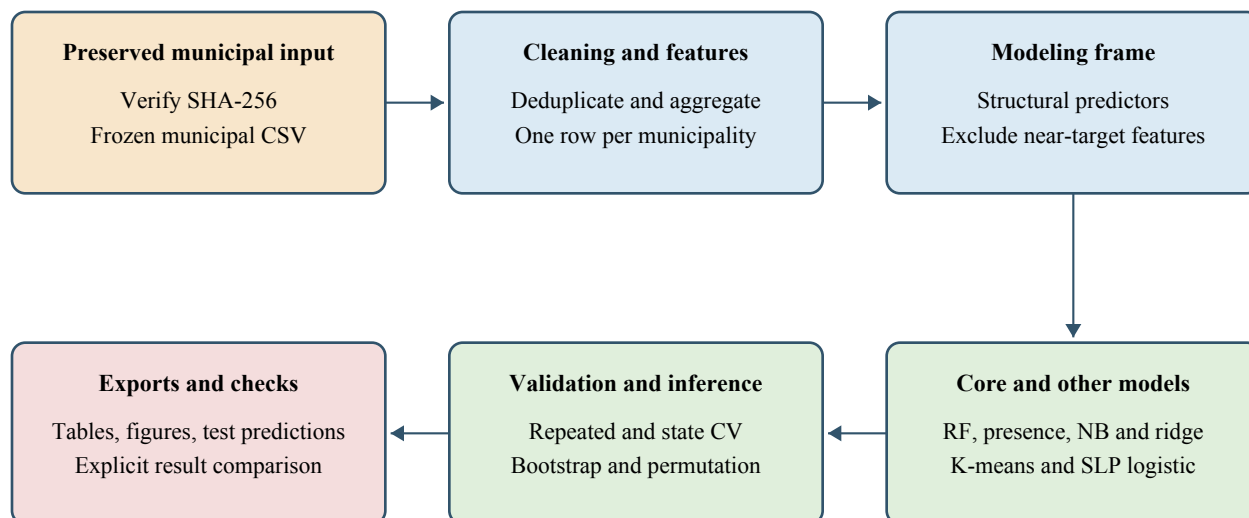
The executable pipeline begins with `data/processed/merged_municipal_dataset.csv`, checks its SHA-256, restores one row per municipality, engineers features and targets, and runs the four research-question analyses. Historical raw-data acquisition is documented separately. Figure 2 records the declared upstream lineage; Figure 10 shows the executable workflow from the preserved input. The workflow creates numerical tables, figures, held-out predictions, split memberships, and diagnostics through `final_pipeline.py`. These outputs describe licensed infrastructure and its municipal correlates. Reproduction uses the source version and pinned environment identified in Appendix A, with input, source, and output hashes connecting the execution to its reported results.

Architecture Diagram

Figure 10

End-to-End Workflow of the Proposed System

Executable workflow from the preserved input



Note. The workflow starts with the preserved municipal CSV, verifies its checksum, and exports analyses and diagnostics. Historical raw acquisition is outside this executable reproduction path.

Workflow Components

Data input: the preserved municipal CSV and its recorded checksum. Historical Anatel/IBGE acquisition is documented separately and is not rerun. Preprocessing removes 193 exact duplicates, collapses 4,343 extra municipal-key rows under explicit rules, aligns names, and selects complete cases; maximum missingness is 0.11%. Outliers remain because the analysis uses transformations rather than deletion. Exploratory outputs cover missingness, skew, observed zeros, correlations, speed by stage, and regional NR presence. Feature engineering constructs fiber and LTE accesses per 100 residents, NR and SLP station intensities per 100,000 residents, log1p variables,

REGION indicators, and percentile targets. Model development uses count/per-capita random forests; presence logistic, truncated and unconditional NB, and supplementary ridge models; k-means with regional association tests; and exploratory SLP classification and inference. Classifiers use stratified splits, ridge uses its regression split, count inference uses its full estimation frame, and clustering is descriptive. Evaluation exports held-out metrics and row-bootstrap intervals, repeated/state CV where implemented, permutation importance, cluster stability, likelihood tests, and clustered coefficient intervals. Held-out municipal predictions and split memberships support audit and exploratory screening. Full-population scoring services, national watchlists, and operational benefits are future applications rather than deployed or experimentally validated outputs. Notebook 06 runs the same pipeline in Colab.

Tools and Technologies

Python 3 with pandas and NumPy for data handling, scikit-learn for machine learning, statsmodels and SciPy for inferential statistics, and Matplotlib for visualization, executed on standard CPU compute; Git and GitHub for version control and public distribution; Google Colab as the zero-setup reproduction environment. No GPU is required.

Results

Model Performance

The results address four research questions. RQ1: structural characteristics discriminate high station counts, with a substantial population-scale signal. RQ2: NR presence is moderately predictable, while full-frame per-capita intensity prediction is limited; density, fiber, and income have modest associations. RQ3: two stable but

weakly separated infrastructure profiles are modestly associated with macro-region. RQ4: higher SLP intensity is associated with wealth and inherited infrastructure, but the finding is exploratory and less stable across states. Tables 11–14 provide the complete predictive comparisons and uncertainty estimates.

Table 11

Model Performance and Inferential Summary

Analysis and estimation frame	Method and scope	Verified results	Interpretation
RQ1 count RF; n = 5,564	Structural predictors; tuned development split	AUC .927 [.906, .945]; F1 .808; accuracy .915	Strong discrimination; substantial municipal-scale signal.
RQ1 per-capita RF; n = 5,564	Separate training-only re-tuning	AUC .817 [.790, .845]; F1 .593; accuracy .809	Above chance, with lower discrimination after normalization.
RQ2 presence; n = 5,564	Logistic classification; development split	AUC .849 [.828, .869]; recall .690; accuracy .796	Moderate discrimination of any licensed NR presence.
RQ2 truncated NB; n = 2,192	Positive counts; population offset; clustered inference	McFadden pseudo-R ² .017; α .65; γ -.139	Modest density, fiber, and GDP associations; underlying mean scaling.
RQ2 unconditional NB; n = 5,564	All counts; population offset; supplementary	McFadden pseudo-R ² .017; α 4.42; γ +.194	Combines entry and intensity; not a hurdle component.
RQ2 full-frame ridge; n = 5,564	Raw/log1p intensity; 1,391 test rows	R ² .02 [-.08, .09] raw; .06 [-.14, .18] log; raw MAE 43.1	Full-frame intervals include zero under the row-bootstrap recipe.
RQ2 conditional ridge; n = 2,192	NR-present log1p intensity; 548 test rows	R ² .114 [.068, .153]; log-scale MAE .608	Modest positive predictive value; conditional on the fitted model.
RQ3 k-means; n = 5,565	Six log1p indicators; k = 2; 20 bootstrap refits	Silhouette .305; ARI .99 $\pm$.01; Cramér's V .21	Stable but softly separated profiles; modest regional association.
RQ4 balanced logistic; n = 5,564	Exploratory SLP; held-out and state checks	AUC .810 [.782, .835]; PR-AUC .623; F1 .576; accuracy .723	Positive wealth and infrastructure associations; weaker state transfer.

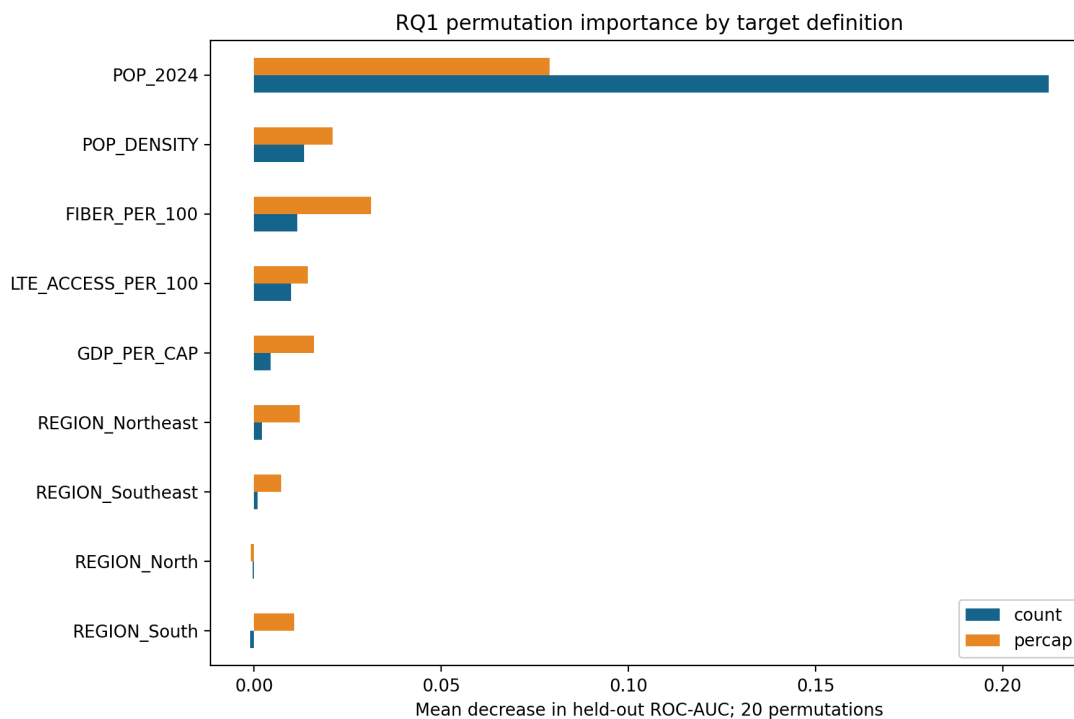
Note. Classifiers use the common frame of 5,564 municipalities; clustering uses 5,565. The truncated NB and conditional ridge use 2,192 NR-present municipalities. Brackets are held-out row-bootstrap 95% intervals (1,000 draws; fitted model fixed). McFadden pseudo- R^2 is likelihood-based and is not interchangeable with held-out predictive R^2 . The conditional ridge target is log-transformed intensity. Tables 12-14 and 19 provide validation, class-reference, and screening details.

Visual Evidence

Figure 11 compares held-out permutation importance for the two RQ1 targets. Permuting POP_2024 reduces count-target ROC-AUC by .212 and per-capita ROC-AUC by .079. Population remains the largest individual importance in both models; its contribution falls while fiber, density, income, and regional indicators gain relative weight. This supports a substantial scale effect without implying that normalization removes population information. Correlated predictors can affect permutation attribution, so the values are descriptive. Table 19 gives the SLP confusion matrix (TP/FN/FP/TN = 262/86/299/744), precision, sensitivity, specificity, and Brier score. Tables 12-14 distinguish training-only CV, full-frame CV, and held-out evaluations. Their differences describe performance sensitivity rather than proving the absence of selection optimism.

Figure 11

RQ1 Held-Out Permutation Importance by Target Definition



Note. Mean ROC-AUC decrease over 20 held-out permutations. Population importance falls from .212 to .079 after normalization but remains the largest individual value; relative contributions of other predictors increase.

Results by Research Question

RQ1: High-Count NR Rollout Classification

Headline finding. Structural characteristics discriminate high-count NR municipalities well, with a substantial scale component. Development ROC-AUC is .927 [.906, .945], repeated-CV AUC is $.920 \pm .009$, and seed-2026 split AUC is .919 [.899, .938]. The intervals support discrimination above chance, conditional on the fitted models. Training-only CV F1 is $.768 \pm .025$ versus held-out .808; this gap does not establish unbiased model selection. Including near-target NR access and speed raises AUC to .965, illustrating why structural-only predictors are retained.

Population is the dominant held-out permutation importance for the count target: permuting POP_2024 reduces ROC-AUC by .212 (Figure 11). When the target is redefined as

NR stations per 100,000 residents, AUC falls to about .82 with count-target settings or separate re-tuning. Repeated CV gives $.824 \pm .017$ and the seed-2026 split gives .825 [.799, .849].

Population importance falls to .079 but remains the largest individual value. Fiber, density, income, LTE, and regional indicators contribute more relative signal. Normalization therefore reduces the dominance of municipal scale without removing it. These comparisons use the same population and do not establish independent prospective performance.

Geographic validation supports the national result with appropriate caution: state-grouped ROC-AUC is $.909 \pm .023$ versus $.919 \pm .003$ under random stratified folds, with greater fold-to-fold variation but only a modest decrease in the mean. Threshold sensitivity at P70 and P80 also preserves the conclusion. Tables 12–14 and Figure 11 provide the full metrics, references, and feature-importance comparison.

Table 12

RQ1 Performance: Naive Versus Leakage-Free Specifications, Per-Capita Redefinition, and Threshold Sensitivity

Specification	Accuracy (majority)	F1 (always-positive)	ROC-AUC [95% CI]	Positive share
Naive (adds near-target NR_ACCESS_PER_100, AVG_DL_SPEED)	.930 (.775)	.835 (.367)	.965	.225
Leakage-free, P75 count target (tuned; training-side CV F1 = $.768 \pm .025$)	.915 (.775)	.808 (.367)	.927 [.906, .945]	.225
Leakage-free, P75 per-capita target, count-target configuration	.813 (.750)	.590 (.400)	.818 [.791, .844]	.250
Leakage-free, P75 per-capita target, re-tuned (training-side CV F1 = .603)	.809 (.750)	.593 (.400)	.817 [.790, .845]	.250
Leakage-free, P70 count target (sensitivity)	.889 (.743)	.768 (.409)	.903	.257
Leakage-free, P80 count target (sensitivity)	.922 (.809)	.778 (.320)	.925	.191

Note. Held-out development metrics; N = 5,564 complete cases. Parentheses give the majority-class accuracy and the always-positive F1 reference for each target; training-side cross-validated F1 is shown in the specification labels so training and test performance can be compared directly. Primary final estimates are reported in Table 14.

Table 13

Random Stratified Versus State-Grouped 5-Fold Cross-Validation (Full Modeling Frame)

Model	CV scheme	F1 (M ± SD)	ROC-AUC (M ± SD)
RQ1 random forest (count target)	Random stratified	.775 ± .014	.919 ± .003
RQ1 random forest (count target)	Grouped by state (27 UFs)	.766 ± .047	.909 ± .023
RQ2 presence logistic	Random stratified	.720 ± .016	.849 ± .014
RQ2 presence logistic	Grouped by state (27 UFs)	.710 ± .047	.839 ± .035
RQ4 balanced logistic	Random stratified	.556 ± .017	.799 ± .017
RQ4 balanced logistic	Grouped by state (27 UFs)	.466 ± .152	.758 ± .067

Note. GroupKFold assigns whole states to folds, so no municipality is validated against neighbours from its own state. N = 5,564. For the RQ2 presence model the random-stratified row is the repeated 5×5 cross-validation estimate of Table 14.

Table 14

Performance Estimates from Repeated CV and Seed-2026 Split Sensitivity

Classifier	Development split, seed 42 (ROC-AUC [95% CI])	Repeated 5×5 CV ROC-AUC (M ± SD; range)	Repeated 5×5 CV F1 (M ± SD)	Fresh split, seed 2026 (ROC-AUC [95% CI]; F1)
RQ1 random forest, count target	.927 [.906, .945]	.920 ± .009; .899–.938	.776 ± .020	.919 [.899, .938]; F1 .773
RQ1 random forest, per-capita target (re-tuned)	.817 [.790, .845]	.824 ± .017; .789–.868	.603 ± .024	.825 [.799, .849]; F1 .590
RQ2 presence logistic	.849 [.828, .869]	.849 ± .014; .821–.878	.720 ± .016	.844 [.821, .865]; F1 .707
RQ4 balanced logistic (exploratory)	.810 [.782, .835]	.798 ± .010; .785–.829	.557 ± .013	.780 [.752, .808]; F1 .534

Note. Repeated CV uses five folds by five repeats on all 5,564 model rows with previously selected settings. These are full-frame validation scores, not training-only CV. Seed 2026 repartitions the same municipalities after earlier tuning and is a split-sensitivity check, not an untouched holdout. Brackets: 1,000 test-row percentile resamples, seed 42 per interval, conditional on the fitted model.

RQ2: Structural Correlates of NR Station Intensity

Headline finding. Presence and positive intensity are distinct targets. Any-NR presence has development AUC = .849 [.828, .869], repeated-CV AUC = .849 ± .014, seed-2026 AUC = .844 [.821, .865], and state-grouped AUC = .839 ± .035. These findings establish moderate discrimination within the evaluated designs; they do not validate a causal account of entry.

Among 2,192 NR-present municipalities, truncated NB is significant at the model level (LLR $p < .001$; McFadden pseudo- $R^2 = .017$). The latter is a likelihood-based fit statistic, not a percentage of variance explained. Per standard deviation, density, fiber, and GDP have underlying rate ratios of 1.25, 1.14, and 1.11, supported by $t(26)$ and state-bootstrap intervals. The centered population coefficient $\gamma = -.139$ gives an underlying count-mean population exponent of .861. The positive-count expectation also includes the truncation adjustment. These coefficients describe associations, rather than causal effects or exact elasticities of observed intensity.

Full-frame ridge gives held-out $R^2 = .02$ (95% CI $[-.08, .09]$) for raw intensity and .06 $[-.14, .18]$ for log1p intensity. H_0 -P is not rejected for these specifications; the intervals provide insufficient evidence of positive full-frame predictive value. The conditional ridge analysis uses 2,192 NR-present municipalities and gives log-intensity $R^2 = .114$ $[-.068, .153]$ (548 test rows; log-scale MAE = .608), supporting modest positive predictive value under the fixed-model row bootstrap. These intervals do not quantify geographic transfer or model-selection uncertainty. Unconditional NB includes zeros; truncated NB estimates positive-count associations.

Table 15

RQ2 Hurdle Part 2: Zero-Truncated Negative-Binomial Model of NR Stations Among NR-Present Municipalities

Predictor	Coefficient (log rate ratio)	Conventional 95% CI	State-clustered, $t(26)$ 95% CI	Cluster-bootstr ap 95% CI	Rate ratio	p (clustered, $t(26)$)
POP_DENSITY (per SD)	+.222	[.171, .274]	[.047, .397]	[.118, .460]	1.25	.015
REGION_South	+.148	[.077, .219]	[.013, .282]	$[-.536, .294]$	1.16	.033

Predictor	Coefficient (log rate ratio)	Conventional 95% CI	State-clustered, t(26) 95% CI	Cluster-bootstr ap 95% CI	Rate ratio	p (clustered, t(26))
FIBER_PER_100 (per SD)	+.134	[.090, .178]	[.069, .199]	[.082, .198]	1.14	< .001
GDP_PER_CAP (per SD)	+.104	[.058, .150]	[.064, .145]	[.052, .150]	1.11	< .001
REGION_Northeast	+.060	[−.012, .131]	[−.067, .187]	[−.691, .198]	1.06	.343
LTE_ACCESS_PER_1 00 (per SD)	+.034	[−.004, .073]	[−.035, .103]	[−.057, .170]	1.03	.314
REGION_Southeast	+.020	[−.050, .089]	[−.083, .123]	[−.674, .206]	1.02	.696
REGION_North	−.019	[−.068, .030]	[−.087, .049]	[−.406, .075]	0.98	.571
log(POP_2024), centred (γ)	−.139	[−.168, −.109]	[−.270, −.008]	[−.244, −.024]	0.87 per log unit	.039

Note. $n = 2,192$; log-population offset; $\alpha = .65$; McFadden pseudo- $R^2 = .017$; LLR $p < .001$. Region indicators are standardized: ratios are per SD, not whole-category contrasts with Center-West. State-pairs bootstrap: 999 draws, 933 retained, 66 rank-deficient; fixed transforms and starts; seed 42. $t(26)$ critical value = 2.056. Population exponent $1 + \gamma$ applies to the underlying mean; the positive-count expectation also adjusts for truncation.

Table 16

RQ2 Supplementary: Unconditional Negative-Binomial Model on the Full Frame

(t(26)-Corrected Clustered Intervals)

Predictor	Coefficient	State-clustered, t(26) 95% CI	p (clustered, t(26))
REGION_South	+.400	[.129, .672]	.005
FIBER_PER_100 (per SD)	+.302	[.140, .463]	.001
REGION_Northeast	+.215	[−.073, .503]	.137
log(POP_2024), centred (γ)	+.194	[.031, .356]	.021
GDP_PER_CAP (per SD)	+.173	[.026, .320]	.023
POP_DENSITY (per SD)	+.149	[.015, .283]	.031
REGION_North	−.040	[−.181, .102]	.569
REGION_Southeast	−.043	[−.277, .190]	.707
LTE_ACCESS_PER_100 (per SD)	−.049	[−.137, .039]	.261

Note. $N = 5,564$; population offset and centered log-population as in Table 15; other predictors and region indicators are standardized. Regional coefficients are per SD, not whole-category contrasts. Including zeros mixes presence and intensity; this is not a hurdle component. Dispersion $\alpha = 4.42$; McFadden pseudo- $R^2 = .017$.

RQ3: Digital-Infrastructure Profile Segmentation

Supporting finding. K-means selects two infrastructure-profile clusters with very high bootstrap stability (adjusted Rand index = $.99 \pm .01$) but only weak-to-moderate separation (silhouette = .305). The higher-profile cluster ($n = 1,231$) has higher average GDP per capita,

density, fiber intensity, NR access, and download speed than the lower-profile cluster. Cluster membership is associated with macro-region, $\chi^2(4) = 254.1$, $p < .001$, but Cramér's $V = .21$ indicates a modest rather than dominant relationship. RQ3 therefore provides supporting evidence of persistent infrastructure geography, not proof that Brazilian municipalities form two natural or sharply distinct types.

Table 17

RQ3 Cluster Profiles (Feature Means by Cluster)

Feature	Higher-profile cluster (n = 1,231)	Lower-profile cluster (n = 4,334)
GDP_PER_CAP (BRL)	47,958.6	29,300.4
POP_DENSITY (per km ²)	419.6	29.9
FIBER_PER_100	155.4	82.8
LTE_ACCESS_PER_100	78.0	76.6
NR_ACCESS_PER_100	29.2	0.7
AVG_DL_SPEED (Mbps)	64.4	38.0

Note. Means on original scales; clustering was performed on standardized log1p features. N = 5,565 complete cases on the clustering feature set. Region $\times$ cluster: $\chi^2(4) = 254.1$, $p < .001$, Cramér's $V = .21$; silhouette = .305.

RQ4: Correlates of High SLP Station Intensity (Exploratory, Supplementary)

Exploratory finding. RQ4 classifies SLP intensity above the 75th percentile.

Development AUC is .810 [.782, .835] and PR-AUC is .623; repeated-CV AUC is $.798 \pm .010$ and seed-2026 AUC is .780 [.752, .808]. Training-only CV F1 is $.549 \pm .016$, compared with held-out .576; full-frame random-fold F1 is separately $.556 \pm .017$. At threshold .5, sensitivity is .753, specificity .713, and precision .467. Accuracy .723 is below the majority reference, reflecting the recall-oriented class weights. State-grouped AUC is $.758 \pm .067$, indicating weaker and more variable geographic transfer.

Coefficient evidence points mainly to wealth and inherited infrastructure: per standard deviation, GDP per capita has OR = 2.62 (t(26) 95% CI [1.39, 4.96]), population OR = 1.68

[1.23, 2.29], LTE intensity OR = 1.45 [1.17, 1.80], and fiber intensity OR = 1.26 [1.09, 1.46].

Three sensitivity analyses address uncertainty in the aggregation rule and analytical threshold.

Maximum, mean, and first-record aggregation preserve the ordering and signs of the main odds ratios; P70 and P80 cutoffs produce ROC-AUC near .80; and the continuous log-intensity model explains 24% of variance with the same broad ordering. These checks bound model dependence but do not verify the historical SLP grain. RQ4 therefore remains a supplementary application concerning private-network-related activity, not private 5G adoption.

Table 18

RQ4 Logistic Regression: Odds Ratios per Standard Deviation With Conventional, State-Clustered, Small-Cluster-Corrected, and Cluster-Bootstrap Intervals

Predictor	β	Odds ratio	Conventional 95% CI	State-clustered 95% CI	t(26)-corrected 95% CI	Cluster-bootstrap rap 95% CI	p (t(26))
GDP_PER_CAP	+.965	2.62	[2.34, 2.94]	[1.43, 4.81]	[1.39, 4.96]	[1.74, 5.37]	.004
POP_2024	+.516	1.68	[1.39, 2.03]	[1.24, 2.26]	[1.23, 2.29]	[1.31, 2.37]	.002
LTE_ACCESS_PER_100	+.373	1.45	[1.29, 1.63]	[1.18, 1.78]	[1.17, 1.80]	[1.20, 1.80]	.001
FIBER_PER_100	+.229	1.26	[1.16, 1.36]	[1.09, 1.45]	[1.09, 1.46]	[1.08, 1.41]	.003
REGION_Southeast	+.027	1.03	[0.92, 1.15]	[0.73, 1.44]	[0.72, 1.46]	[0.49, 1.40]	.878
POP_DENSITY	-.052	0.95	[0.88, 1.02]	[0.89, 1.01]	[0.89, 1.01]	[0.83, 1.08]	.115
REGION_North	-.172	0.84	[0.77, 0.92]	[0.74, 0.95]	[0.74, 0.96]	[0.68, 0.93]	.013
REGION_South	-.405	0.67	[0.60, 0.74]	[0.45, 0.99]	[0.44, 1.01]	[0.39, 1.07]	.055
REGION_Northeast	-.548	0.58	[0.51, 0.66]	[0.44, 0.76]	[0.44, 0.77]	[0.41, 0.73]	< .001

Note. N = 5,564; all predictors standardized, including region indicators. Regional ORs are per SD, not whole-category contrasts. LLR $p < .001$; McFadden pseudo- $R^2 = .196$; $t(26) = 2.056$. Of 999 state draws, 898 were retained; 66 were rank-deficient and 35 nonconverged/nonfinite. Fixed transforms/starts; seed 42.

Table 19

RQ4 Held-Out Screening Metrics at the Default Threshold (.5)

Metric	Value	Reference
ROC-AUC [bootstrap 95% CI]	.810 [.782, .835]	.50 (chance)
PR-AUC (average precision)	.623	.250 (prevalence)
Sensitivity (recall)	.753	—

Metric	Value	Reference
Specificity	.713	—
Precision	.467	.250 (prevalence)
F1	.576	.400 (always-positive)
Accuracy	.723	.750 (majority class)
Brier score	.176	.188 (predicting prevalence)
Confusion (TP / FN / FP / TN)	262 / 86 / 299 / 744	test n = 1,391

Note. Class-weighted model; the .5 threshold favours recall over specificity by design. The Brier reference is the score of a constant prediction equal to the prevalence, $p(1 - p)$.

Table 20

RQ4 Sensitivity: Aggregation Rule for the Two SLP Records per Municipality and Cut-Off

Percentile

Specification	Threshold (per 100,000)	Agreement with baseline target	Held-out ROC-AUC	OR GDP per capita	OR population	OR LTE	OR fiber	OR Northeast
Sum (baseline), P75	1,618	100%	.810	2.62	1.68	1.45	1.26	0.58
Maximum, P75	1,283	97.1%	.777	2.49	1.53	1.41	1.22	0.57
Mean, P75	903	97.3%	.789	2.64	1.53	1.48	1.18	0.57
First record, P75	1,126	94.6%	.779	2.21	1.63	1.45	1.22	0.55
Sum, P70	1,272	—	.797	2.60	1.98	1.57	1.24	0.58
Sum, P80	2,268	—	.801	2.39	1.52	1.30	1.24	0.56

Note. Full-frame logistic point estimates; all predictors, including region indicators, standardized. The Northeast OR is per SD of its indicator. Agreement measures shared target labels with the sum baseline. The continuous complement uses $\log(1 + \text{SLP per } 100,000)$: $R^2 = .244$ with state-clustered $t(26)$ inference.

Overall Interpretation

The central finding is strong discrimination of high station counts alongside limited continuous intensity prediction across the full municipal modeling frame. The count RF attains repeated-CV AUC near .92; its per-capita counterpart is near .82, and NR presence is moderately predictable. Full-frame ridge R^2 is .02 for raw intensity and .06 for log intensity, with both 95% intervals including zero. These results support screening applications, while evidence for full-frame intensity forecasting remains limited. The inferential analysis identifies

modest density, fiber, and GDP associations. The population coefficient indicates sub-proportional scaling of the underlying count-mean parameter; interpretation of the positive-count expectation also requires the truncation adjustment.

The broader evidence is consistent with the infrastructure-economics and Brazilian digital-divide literature. Population scale and accumulated infrastructure shape entry; regional inequality remains visible, but the profile clusters are a soft partition and region is not destiny. The exploratory SLP model similarly links intensity to wealth and existing infrastructure, yet its weaker state-level generalization and unresolved historical grain prevent it from carrying the same evidentiary weight as RQ1–RQ3.

The study contributes a reproducible municipal screening framework that connects structural indicators to licensed NR presence, high-count rollout, and infrastructure profiles. Its national-frame findings support investigation and benchmarking, with limited performance for continuous per-capita intensity prediction. The evaluation accounts for population scale, excludes near-target variables from structural models, and reports geographic sensitivity and uncertainty. The outputs provide evidence for assessing potential screening uses; they do not establish causation, certify Standalone or cloud-native operation, or determine regulatory compliance. Engineering and investment decisions require operator, spectrum, cost, demand, and site-level information that municipal aggregates do not contain.

Practical Significance

For planning, RQ1 scores offer screening information, RQ3 profiles offer descriptive benchmarks, and RQ4 offers an exploratory SLP signal. The classification and profile results can help prioritize investigation; full-frame intensity forecasts remain limited. These proposed applications require external validation before operational use, together with

current inputs and site-level cost, spectrum, and demand evidence. Structural municipal scores alone are insufficient to establish engineering or investment feasibility.

Implementation and User Benefit

Deployment Approach

The analysis package contains `final_pipeline.py`, shared analysis modules, and a Colab entry point. It generates numerical tables, figures, held-out predictions, and a machine-readable results file on CPU compute. A local environment requires the pinned dependencies; the Colab notebook installs them before execution. A future scoring service could refit on authenticated, dated source updates and produce national rankings, subject to new validation. The input checksum verifies the dataset used for this study. The generated artifacts support model evaluation and reproducibility; deployment as a scoring service would require additional validation and implementation.

System Integration

The executable input is the preserved merged CSV. Held-out prediction and split-membership exports carry seven-digit IBGE municipal identifiers, allowing linkage to GIS or planning data for evaluation. These exports cover the declared test partitions; they are not full-population, independently validated probability rankings. A future carrier application could combine externally validated scores with cost, spectrum, and demand information. Regulatory use would require coverage-obligation data, which this study does not observe. The current analysis supplies structural context rather than a compliance or deployment-status determination.

User Interaction

Analysts can run the pipeline, inspect the generated tables and figures, and audit municipality-level held-out predictions. Planners can review these evaluation results alongside the cluster profiles and the exploratory SLP metrics. National watchlists, municipality-level cluster exports, a user interface, and full-frame probability rankings remain implementation steps. Before operational use, analysts would need to validate the intended scoring population, calibrate class-weighted probabilities, select thresholds for the use case, and assess performance on independent or future data.

Benefits to Users

The intended benefit is a reproducible starting point for comparing municipal infrastructure and prioritizing investigation. Such screening could reduce survey or planning effort, but this study did not measure those savings or validate a production shortlist. Profile summaries let analysts compare broad economic and infrastructure conditions; they do not certify site-level feasibility. The data omit deployment cost, operator strategy, spectrum holdings, industrial composition, and enterprise counts. Consequently, proposed benefits require operational testing, and the evidence cannot independently justify capital allocation, private 5G adoption, edge computing, or AI-RAN decisions.

Example Use Case

A future planning tool could join municipality-level cluster assignments to validated rollout and SLP scores, then identify higher-income, fiber-rich areas for investigation. The higher-profile cluster contains 1,231 municipalities, but a nationwide joined shortlist is not an output validated in this study. Any proposed private-network review would begin with site-level cost and demand

analysis rather than a model-only decision. A regulator could likewise investigate mismatches between validated structural scores and recorded stations, after linking the relevant obligation and coverage evidence.

Limitations and Further Improvements

Limitations

Six limitations bound the findings. First, construct validity is indirect: licensed NR stations do not establish 5G Standalone, cloud-native core, or virtualized RAN operation, and SLP stations do not identify private 5G. Second, the primary count target partly encodes municipal scale; the per-capita model quantifies but does not eliminate the fact that the two targets answer different questions. Third, the cross-sectional predictors are structural correlates rather than exogenous causes, and several source reference periods are not recorded precisely enough to establish temporal ordering.

Fourth, validation and uncertainty remain limited. State transfer is weaker for SLP; fixed-configuration CV and the seed-2026 repartition are not nested or untouched evaluations. Ridge scaling precedes inner CV, and test-row intervals omit model-selection and spatial uncertainty. With only 27 states, clustered intervals and filtered state bootstraps remain approximate. Fifth, historical raw snapshots are unavailable; SLP grain and measured-speed provenance remain unresolved. Reproduction is verified from the preserved merge, not authenticated historical reacquisition.

Sixth, analytical thresholds and clustering simplify continuous phenomena. P70/P80 and continuous-outcome sensitivity checks reduce dependence on the selected quartile, but “high” means above this study's cutoff rather than an industry or regulatory standard. The RQ3 silhouette indicates soft separation, and the literature base remains stronger on general

telecommunications economics and digital inequality than on recent Brazilian municipal 5G investment behavior.

Impact of Limitations

Count AUC should be interpreted alongside the per-capita result, coefficients remain associational, and SLP findings remain exploratory. Full-frame intensity models have low held-out R^2 and confidence intervals that include zero. These results indicate limited forecasting performance with the available structural data. They do not demonstrate the absence of every predictive relationship. Source links identify datasets without authenticating historical snapshots. Engineering sufficiency and transfer to future data require independent evaluation.

Future Improvements

The highest-value next step is to rebuild the merge from newly downloaded, dated raw extracts and archive each URL, retrieval timestamp, schema, and SHA-256 checksum locally; comparing those files with any surviving historical raw files would establish the precise vintage and could confirm the SLP aggregation grain. Methodologically, a longitudinal panel around auction-obligation dates, state fixed effects or multilevel models, contiguity-blocked validation, and explicit spatial models (Anselin, 1988) would move the design beyond cross-sectional association. Construct validity would improve through fields that distinguish SA from NSA, radio virtualization, spectrum band, operator strategy, industrial composition, enterprise counts, deployment cost, and verified private-network technology. A systematic Brazilian and Latin American literature search and wild-cluster or CR2 inference would further strengthen the scholarly and statistical foundation.

Future Scope

The framework extends naturally to a longitudinal panel as Anatel updates accumulate, enabling designs that observe change around obligation deadlines rather than a single cross-section; to other countries with comparable open-data infrastructures; and, once such registries become public, to outcome variables that measure the constructs directly — standalone-core status, virtualized RAN deployment, and private 5G licences — rather than their radio-station proxies.

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Appendix A Analysis and Computational Reproducibility

The analysis was executed on September 14, 2026. The project repository is <https://github.com/ronyspada2025/walsh-msc-capstone>. The source package `walsh_capstone_update.zip` contains the executable analysis and generated outputs. The execution identifier and recorded source, input, and artifact hashes identify the version supporting the numerical results in this report.

Execution identifier: 5dcc6e28649f4cc385ff134e8db4745d. Complete run on Linux; Python 3.12.14.
Input: `data/processed/merged_municipal_dataset.csv`.

Input SHA-256: 50fac84b16f63d66628741f36686cc076f43c90632a3befe53ca43ed9b316207

Numerical packages: NumPy 2.3.5; pandas 2.2.3; SciPy 1.17.0; scikit-learn 1.8.0; statsmodels 0.15.0; Matplotlib 3.10.8. `requirements.txt` locks the dependency closure. The analysis uses CPU compute; no GPU is required.

Reproduce and verify

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
.venv/bin/python final_pipeline.py --jobs 4
.venv/bin/python scripts/verify_project.py
```

For comparable resource use, set `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, and `MKL_NUM_THREADS` to 1 and `LOKY_MAX_CPU_COUNT` to 4 before execution. Notebook 06 invokes the same entry point in Colab; notebooks 01-05 inspect shared preparation or generated outputs.

Source modules integrate preparation, predictive validation, count and SLP inference, report exports, and integrity checks. `reports/run_manifest.json` records generated artifact hashes; `reports/tables/headline_results.json` records source hashes, software versions, warnings, and numerical results. Appendix B documents the official data sources and historical acquisition limitations.

Appendix A Estimation and Validation Details

The RF grid crosses 300/600 trees, depth None/8/14, and minimum leaf size 1/5, with balanced class weights. Count and per-capita targets are tuned separately on seed-42 training rows. Seed-2026 partitions and repeated 5×5 CV use previously selected settings; they are sensitivity analyses, not nested or untouched assessments of model selection.

Held-out percentile intervals use 1,000 row resamples, restarting RandomState(42) for each statistic. They hold the fitted model fixed. Ridge uses StandardScaler followed by RidgeCV over 30 penalties with five folds; scaling precedes those inner splits. The conditional ridge target is log1p NR intensity among NR-present rows.

Truncated NB uses positive counts, a log-population offset, centered log population, and other predictors standardized within its estimation frame. State-pairs bootstrap draws 27 states with replacement 999 times, seed 42, retaining fixed transforms and full-fit starts. Rank-deficient, nonconverged, and nonfinite fits are excluded and recorded: 933 usable truncated-NB draws and 898 SLP draws. The t(26) intervals and bootstrap intervals express different approximations.

Exported evidence includes held_out_predictions.csv, split_membership.csv, ridge_held_out_predictions.csv, repeated_cv_fold_scores.csv, count/SLP bootstrap coefficient draws and diagnostics, report-numbered tables, and generated figures. These files document fitted-model evaluation and the analysis partitions.

Reproducibility checks

scripts/verify_project.py checks input, source and artifact integrity, completeness, partitions, convergence, and implementation benchmarks. Eight automated implementation checks and a fresh-clone update test passed. The shell verification wrapper checks existing outputs; it does not rerun every notebook.

Computational reproducibility is established for the preserved municipal input and recorded software environment. Validation remains limited to the specified municipal splits, fixed-configuration repeated and state-grouped checks, and bootstrap procedures. Independent future-data evaluation, authenticated historical raw-data reacquisition, and production deployment remain outside this study.

Appendix B: Source Data Provenance and Original Public Links

This appendix documents the public source path for every source family used in the frozen municipal merge. Class A identifies the exact dataset/table and vintage at the dataset level; Class B identifies the official family and current resource but lacks the original snapshot bytes or complete reference period; Class C identifies the official family but retains unresolved historical schema or grain. In every class, the surviving links support source identification, not a claim that a current download is byte-identical to the historical extract. Where only a resource file name survives (B6–B8), that name is stated explicitly as a candidate rather than presented as a full URL. The accompanying CSV manifest (docs/data_provenance_manifest_v6.csv) contains the same information in machine-readable form. All landing pages and table identifiers below were verified as resolving on July 23, 2026.

B1. IBGE_CENSUS_2022 — Census 2022 municipal aggregates / SIDRA

Publisher: IBGE

Official landing page: <https://sidra.ibge.gov.br/>

Current direct resource or table identifier: <https://sidra.ibge.gov.br/tabela/4714>

Variables used: POP_CENSUS_2022; AREA_KM2; POP_DENSITY

Reference-period status: 2022 recorded

Historical snapshot status: original downloaded bytes not archived

Grain status: 1 row per municipality

Assurance class: A

Analysis use: RQ1-RQ4 structural context

Verification gate: Confirm local extract columns correspond to Table 4714 variables; archive checksum in next dated release

Limitation: Class A is dataset-level identification, not byte-identical historical reproduction

B2. IBGE_POP_2024 — Annual municipal population estimate / SIDRA

Publisher: IBGE

Official landing page: <https://sidra.ibge.gov.br/>

Current direct resource or table identifier: <https://sidra.ibge.gov.br/tabela/6579>

Variables used: POP_2024

Reference-period status: 2024 recorded

Historical snapshot status: original downloaded bytes not archived

Grain status: 1 row per municipality

Assurance class: A

Analysis use: RQ1-RQ4 predictor

Verification gate: Match local population file or table export to Table 6579; record checksum

Limitation: Dataset and vintage identified; original export bytes absent

B3. IBGE_GDP_MUNICIPAL — Produto Interno Bruto dos Municípios / SIDRA

Publisher: IBGE

Official landing page:

<https://www.ibge.gov.br/estatisticas/economicas/contas-nacionais/9088-produto-interno-bruto-dos-municipios.html>

Current direct resource or table identifier: <https://sidra.ibge.gov.br/tabela/5938>

Variables used: PIB_MIL_REAIS; GDP_PER_CAP

Reference-period status: original extract year unrecorded

Historical snapshot status: original downloaded bytes not archived

Grain status: 1 row per municipality

Assurance class: B

Analysis use: RQ1-RQ4 predictor

Verification gate: Identify the local GDP file and its reference year; compare schema to Table 5938; record checksum

Limitation: GDP_PER_CAP combines GDP of an unrecorded historical year with POP_2024

B4. ANATEL_SMP — Mobile accesses by technology / Acessos Autorizadas SMP

Publisher: Anatel

Official landing page: <https://informacoes.anatel.gov.br/paineis/>

Current direct resource or table identifier:

https://www.anatel.gov.br/dadosabertos/paineis_de_dados/acessos/acessos_telefonia_movel.zip

Variables used: ACC_CDMA_IS_95; ACC_GSM; ACC_LTE; ACC_NR; ACC_WCDMA; LTE_ACCESS_PER_100; NR_ACCESS_PER_100

Reference-period status: monthly; original reference month unrecorded

Historical snapshot status: original August 2026 ZIP/checksum not archived

Grain status: provider-period records aggregated to municipality

Assurance class: B

Analysis use: LTE structural predictor; NR access only RQ3 and leakage demonstration

Verification gate: Match local filename/header to current ZIP; identify reference month and aggregation fields

Limitation: Current direct ZIP does not prove that the historical file is byte-identical

B5. ANATEL_ERB — Licensed radio stations / Estacoes Licenciadas

Publisher: Anatel

Official landing page: <https://www.anatel.gov.br/dadosabertos/>

Current direct resource or table identifier:

https://www.anatel.gov.br/dadosabertos/paineis_de_dados/outorga_e_licenciamento/estacoes_licenciadas.zip

Variables used: ERB_CDMA; ERB_EDGE; ERB_GSM; ERB_LTE; ERB_NR; ERB_WCDMA; NR_STATION_CNT

Reference-period status: continuous registry; historical snapshot date unrecorded

Historical snapshot status: original August 2026 ZIP/checksum not archived

Grain status: station/technology records aggregated to municipality

Assurance class: B

Analysis use: RQ1-RQ2 outcome

Verification gate: Match local filename/header and station-count logic to current ZIP; recover snapshot date if possible

Limitation: Licensed NR station count is not evidence of SA-core or cloud-native operation

B6. ANATEL_FIBER — Meu Municipio / fiber and backhaul panel

Publisher: Anatel

Official landing page: <https://informacoes.anatel.gov.br/paineis/>

Current direct resource or table identifier: resource file name candidate only — meu_municipio.zip (no full historical URL survives; the panel landing page above is the verified access point)

Variables used: FIBER_ACCESSES; FIBER_BACKHAUL; FIBER_PER_100

Reference-period status: periodic; original reference period unrecorded

Historical snapshot status: original ZIP/checksum not archived

Grain status: municipality-level after aggregation

Assurance class: B

Analysis use: RQ1-RQ4 structural predictor

Verification gate: Locate local meu_municipio file; compare member names/header and municipal key; record period/checksum

Limitation: Fiber is prior infrastructure context and not an exogenous causal variable

B7. ANATEL_SLP — Redes Privativas / Servico Limitado Privado

Publisher: Anatel

Official landing page: <https://www.gov.br/anatel/pt-br/regulado/radiofrequencia/redes-privativas>

Current direct resource or table identifier: resource file name candidate only — redes_privativas.zip; the documented public data panel for these authorizations is Atos de RF – Redes Privativas at <https://informacoes.anatel.gov.br/paineis/>

Variables used: SLP_STATION_CNT; PRIVATE_5G_LIC; SLP_PER_100K_POP; HIGH_SLP

Reference-period status: continuous registry; historical snapshot date unrecorded

Historical snapshot status: original ZIP/checksum not archived

Grain status: two records for 4,343 municipalities in frozen merge; source meaning unresolved

Assurance class: C

Analysis use: RQ4 exploratory only

Verification gate: Compare local columns and multiplicity with the current resource and registry documentation; establish whether sum/max/mean/first has a documented meaning

Limitation: Sensitivity bounds model dependence but does not verify the historical record grain or private-5G adoption

B8. ANATEL_SPEED — Measured mobile speed / download-map panels

Publisher: Anatel

Official landing page: <https://informacoes.anatel.gov.br/paineis/>

Current direct resource or table identifier: resource file name candidates only — medidas_qoe_smp.zip; alternative schema candidate mapa_download_smp.zip (no full historical URL survives)

Variables used: AVG_DL_SPEED

Reference-period status: periodic; original reference period unrecorded

Historical snapshot status: original ZIP/checksum not archived

Grain status: municipality-level mean in frozen merge; source schema unresolved

Assurance class: C

Analysis use: EDA and RQ3 only

Verification gate: Compare each local speed file header and aggregation fields against both candidates; document exact source

Limitation: Excluded from RQ1-RQ2; a plausible resource name alone is insufficient for attribution